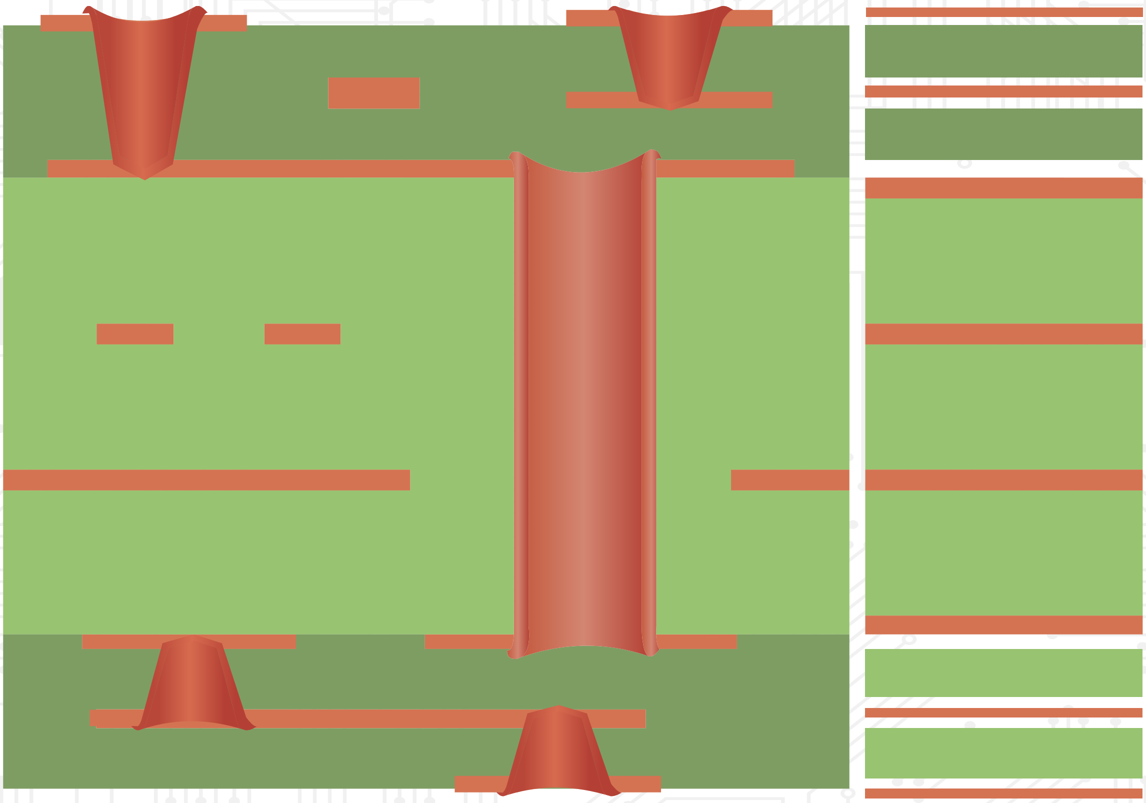
PCB Via Design Guide



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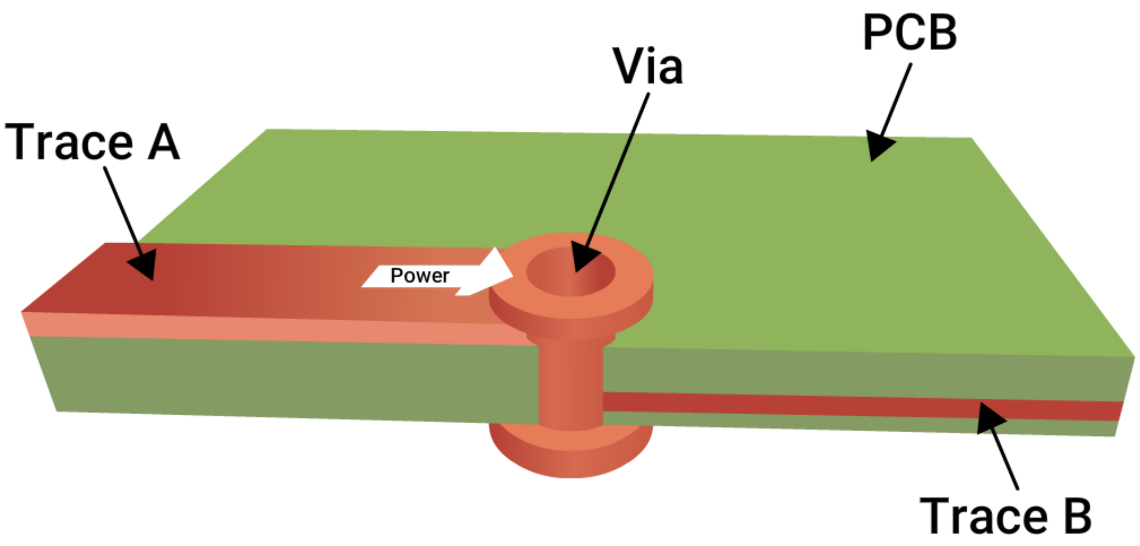
1. Introduction to PCB Vias

PCBs are built with multiple stacked copper layers to support complex routing, efficient power distribution, and high component density. To connect these layers electrically, designers use vias, which are small plated holes filled with conductive or non-conductive material.

1.1. Functionality of vias: what do they really do?

Vias act as vertical pathways that allow electrical signals or power to transition between different layers of a printed board.



Current transition through a via

Without vias, signal routing would be restricted to a single copper layer, severely limiting board density and functionality. Vias can connect components to internal power planes, link signal traces between layers, or provide heat dissipation paths for thermal management.

Vias influence several key performance characteristics:

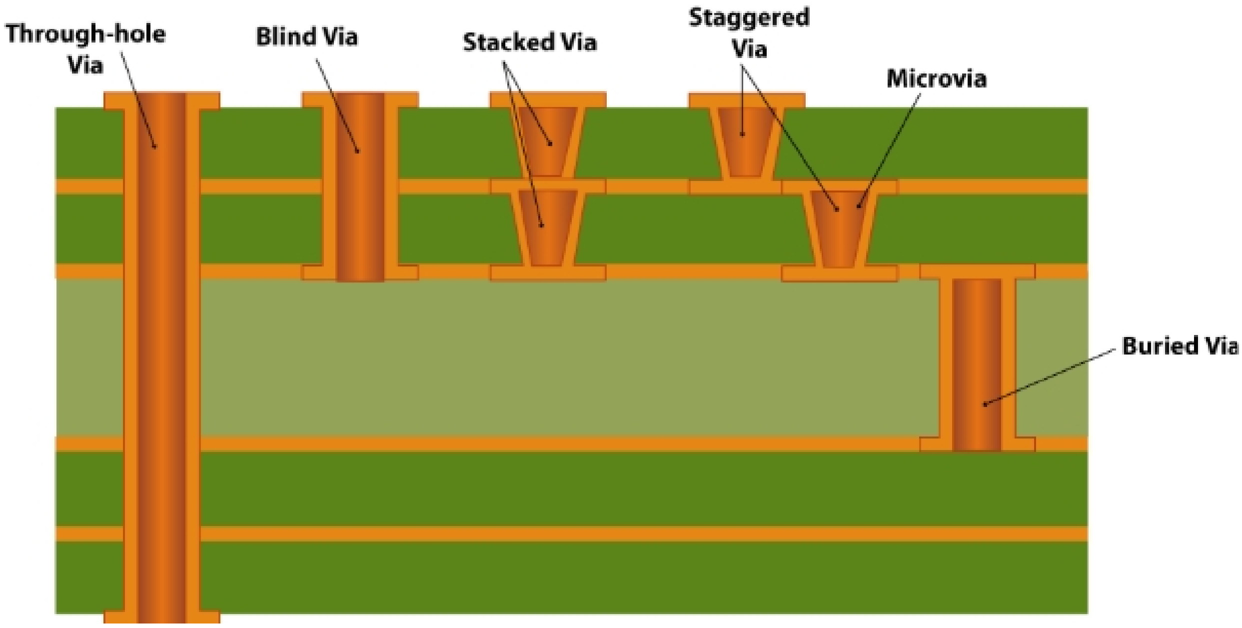
- Signal integrity (SI): Poorly placed or stubby vias can cause reflections, impedance mismatches, and signal degradation at high frequencies.
- Power integrity (PI): Vias serve as power delivery paths and should be designed to handle the desired current and reduce IR drop.

- Thermal management: Thermal vias dissipate heat from power components to internal or bottom layers with ground planes.

From a mechanical perspective, vias must withstand thermal cycling, mechanical stresses, and fabrication tolerances. Poorly designed vias can result in barrel cracking, plating voids, or delamination, especially in class 3 applications.

1.2 When and where to use through-hole, blind, buried, and microvias

The choice of via type depends on 3 major factors: how a PCB is constructed, where the signal needs to travel, and the desired board complexity.



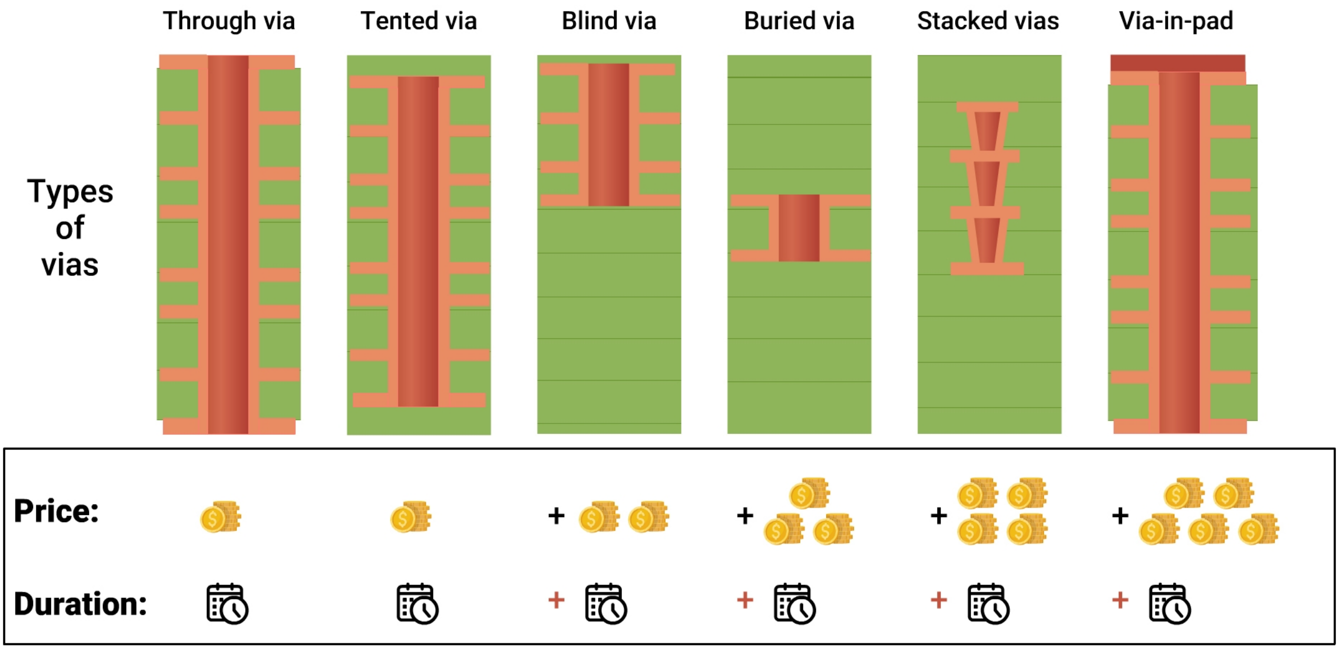
Different types of vias

1.2.1 Comparison between via types and use cases

Via type	Description	Used in	Pros	Cons
Through-hole	Drilled from top to bottom of the PCB	2+ layer standard boards	Cost-effective and easy to manufacture	Consumes space on all layers causes stub effects
Blind via	Connects the outer layer to one or more inner layers	HDI boards and compact multilayer designs	Saves space in inner layers and reduces the risk of stubs	Requires sequential lamination and costs more
Buried via	Connects internal layers only	High-density or RF boards	Doesn't affect outer layers, which is good for RF designs	Invisible for inspection adds to the complexity

Via type	Description	Used in	Pros	Cons
Microvia	Laser-drilled, small-diameter via placed between two adjacent layers	HDI, smartphones, fine-pitch BGAs	Allows tight routing, ideal for a fine pitch with lower parasitics	Limited depth (one layer at a time). It is comes with additional cost

Microvia, blind, and buried vias require separate lamination cycles, adding complexity to the stack-up and affecting both manufacturing yield and overall cost.



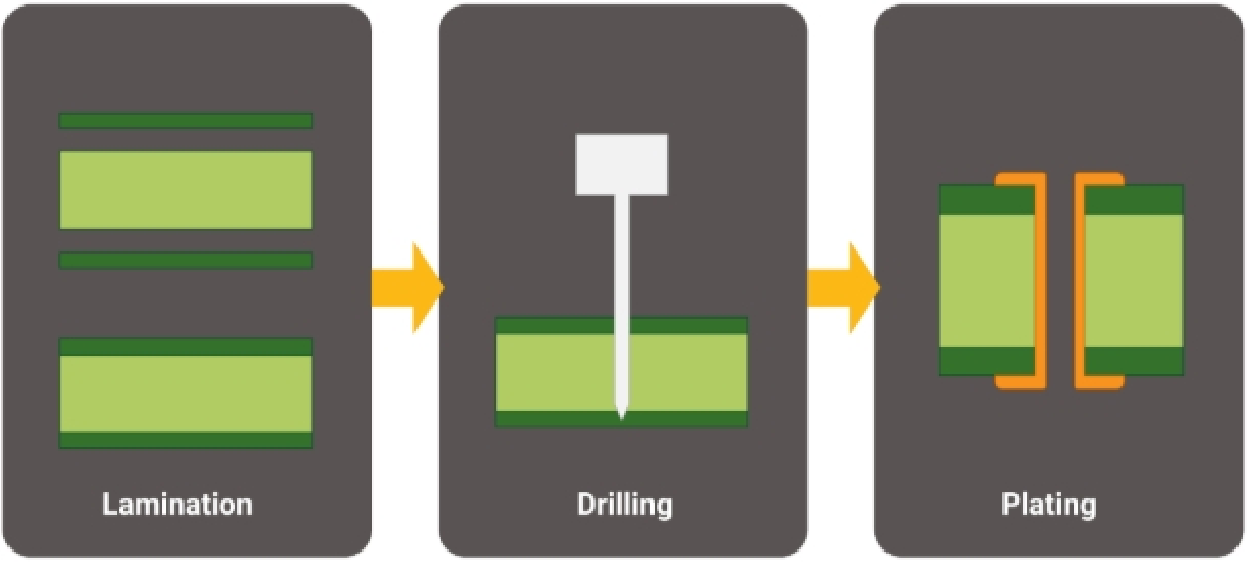
Types of drilled holes vs manufacturing cost

1.2.2 Common PCB design scenarios and via recommendations

The table below outlines common PCB design scenarios along with the most suitable via types.

Design scenario	Recommended via type	Reason
Fine-pitch BGA fanout	Microvia or via-in-pad	Saves space and supports high-speed signal routing
Power delivery to an internal power/ground plane	Through-hole	Offers better current-carrying capacity
RF/microwave signal routing	Blind or buried via	Maintains controlled impedance and clean return paths
Thermal management for high-power components	Array of thermal vias	Dissipates heat towards the inner copper layers
High-density mobile or wearable designs	Microvia	Enables ultra-dense routing and reduces layer count

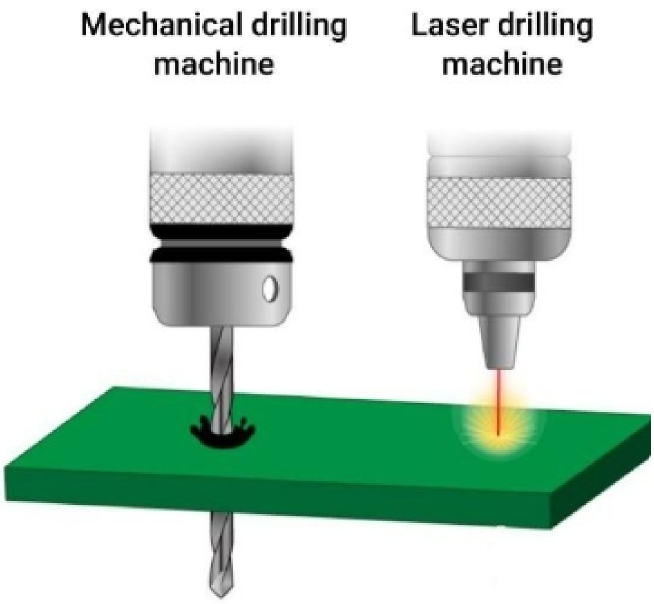
1.3 How are vias manufactured?



Via manufacturing process

For PCB designers, understanding the via fabrication process is essential to ensure DFM compliance, maintain signal integrity, and enhance overall board reliability. Among the various steps involved, drilling and plating (metallization) are the two most critical stages in via formation.

1.3.1 Drilling techniques and their impact on via reliability



Mechanical and laser drilling machines

Via formation begins with drilling, creating a physical hole that will later become an electrical interconnect. The chosen drilling method, via size and precision, impacts both manufacturability and performance.

The table below compares common via drilling methods, highlighting their size ranges, ideal use cases, advantages, and limitations.

Comparison of common via drilling methods

Method	Typical hole size	Use case	Advantages	Limitations
Mechanical drilling	8 - 32 mil	Through-hole and buried vias	Cost-effective and easy to manufacture	Tool wear, minimum via size >6 mil
Laser drilling	3 - 6 mil	Blind vias and microvias (HDI)	Saves space in inner layers and reduces the risk of stubs	Higher cost and risk of resin burning
Plasma etching	<4 mil	Flex and rigid-flex PCB via formation	Uniform removal for fine vias in flex	Slower process and requires specialized equipment

1.3.2 Plating and metallization processes

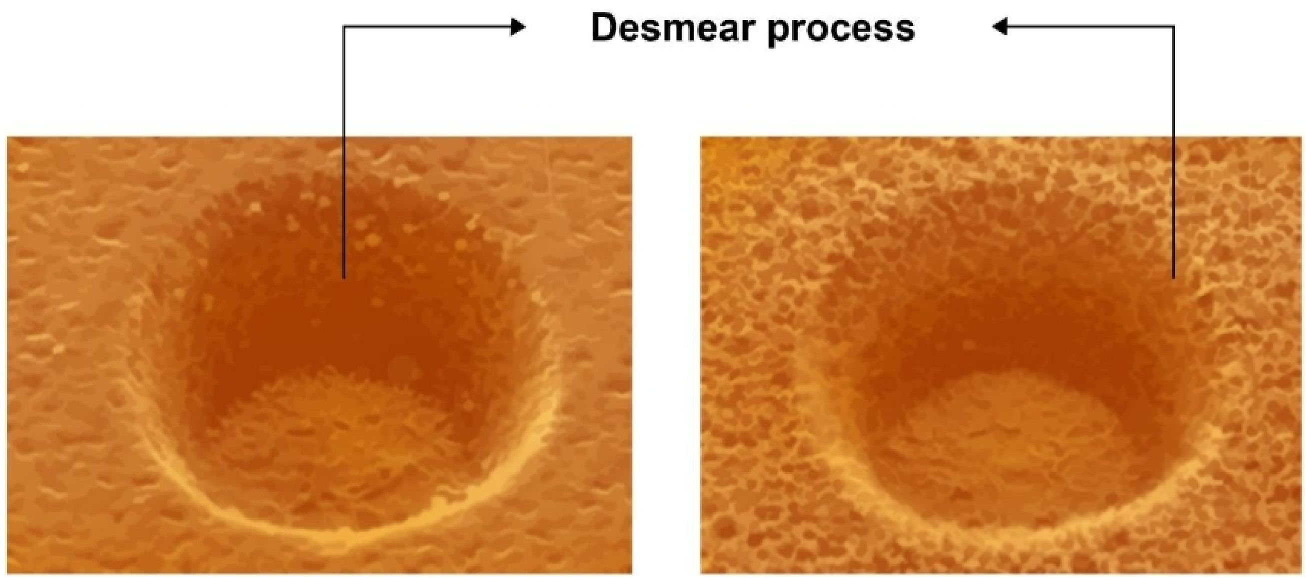
Once drilled, vias must be metallized to form an electrical pathway. This is a multi-step process that determines the conductivity, mechanical strength, and thermal cycling durability of the via.

The steps below describe the metallization process:

1. Desmear and conditioning

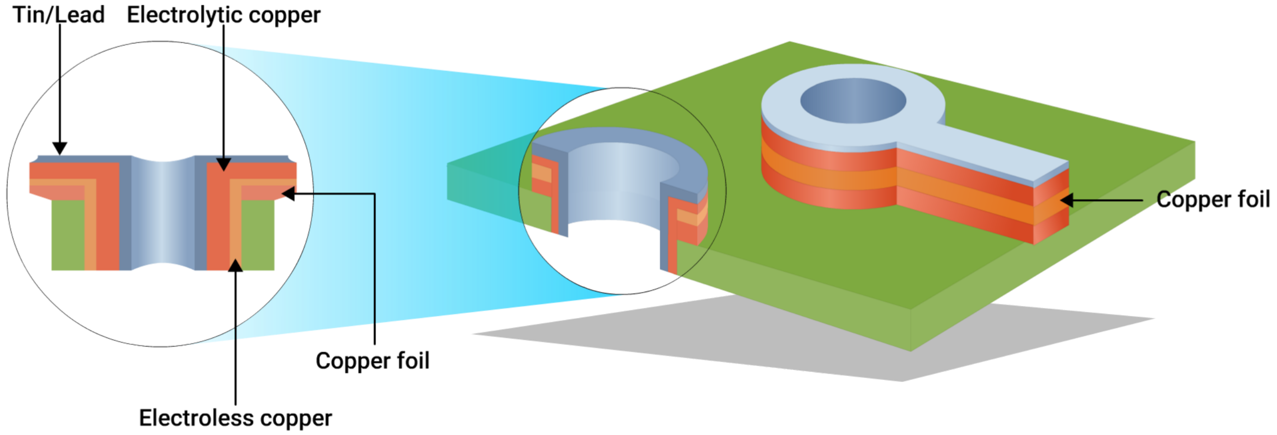
Drilling, especially laser or mechanical, can leave resin smear on the inner copper layers. This smear impairs plating adhesion.

- Chemical desmear: Uses $KMnO_4$ + H_2SO_4 to etch the resin.
- Plasma desmear: Oxygen plasma removes residue with uniform surface conditioning.



Holes after the desmear process

2. Electroless copper deposition



Electroless copper deposition process

This step coats the via walls with a thin, conductive copper layer (1–3 µm) through a chemical reaction, creating a base for electroplating.

This is essential for ensuring uniform copper coverage in non-conductive areas (via barrels). Poor deposition leads to voids, especially critical in class 3 boards.

3. Electrolytic copper plating

Once seeded, copper is thickened using DC current in an acid copper bath. This forms the main current-carrying barrel of the via.

Thickness variation or insufficient coverage can cause open vias, especially under thermal stress. Throwing power (plating thickness at hole center vs surface) is a key quality metric.

Minimum barrel plating thickness required as per class 2 is 20 µm (0.8 mil), and class 3 is 25 µm (1 mil). Plating must wrap around via barrels and connect inner layers securely.

Proper via manufacturing ensures board longevity, especially for HDI, RF, or aerospace designs. Even the best layout can fail if vias are not drilled or plated correctly.

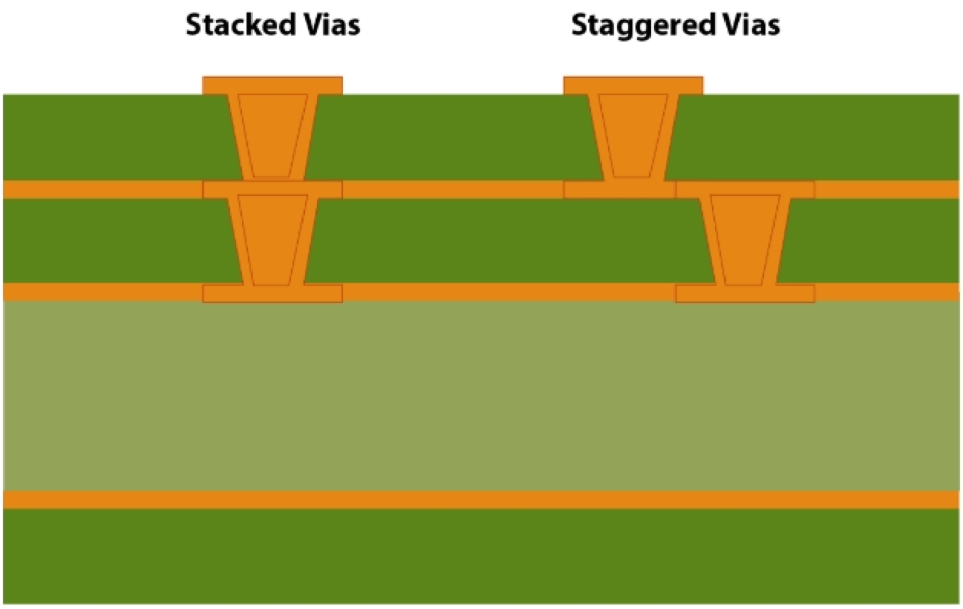
2. Advanced via structures and their applications

Advanced via structures are engineered to meet the demands of high-density and high-speed PCB designs. In this section, you'll learn key technologies including stacked and staggered vias, via-in-pad designs, via stitching, and thermal vias.

2.1 Staggered and stacked vias

As PCB designs advance to accommodate high I/O densities and fin pitch components, laser-drilled microvias have become critical for maintaining signal integrity and meeting routing constraints.

Two widely used configurations, staggered and stacked microvias, offer distinct advantages depending on design priorities. These include routing density, signal performance, fabrication complexity, and long-term reliability.



An illustration of stacked and staggered vias

Staggered microvias:

- Are placed offset from one another across layers.
- Connect through layers in a step-like fashion.
- Avoid overlapping drill sites.

Stacked microvias:

- Are aligned directly one above another.
- Create a continuous via through multiple layers.

- Can be blind or buried microvias aligned end-to-end.

According to IPC-T-50M, microvias are defined as vias with a diameter of 0.010 inches (254 µm) or less and an aspect ratio of 1:1 or lower. While stacked microvias are common in HDI designs, using more than two stacked layers can reduce reliability.

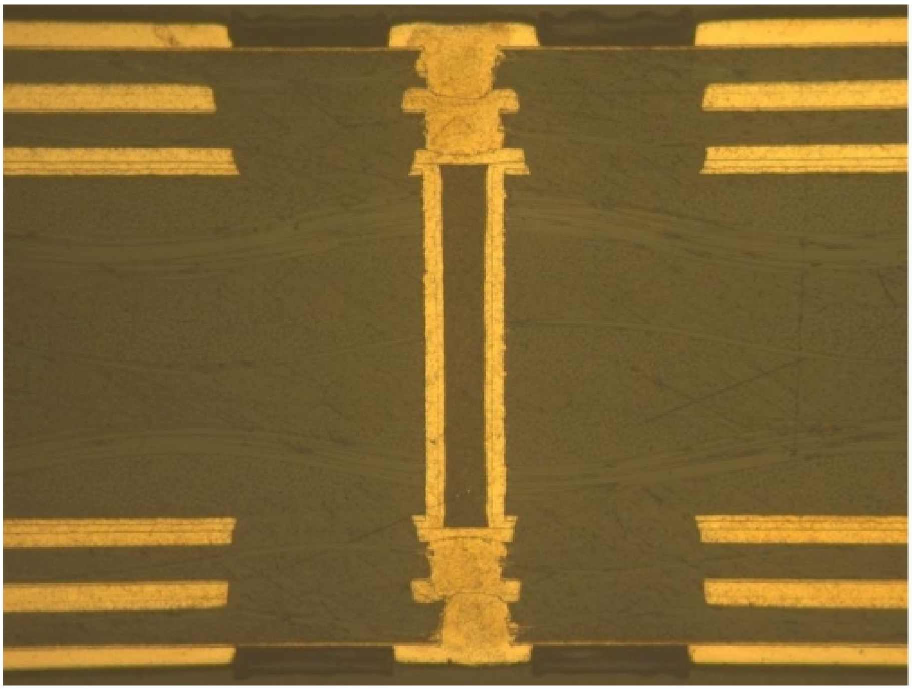
The table below compares different microvia structures based on their drill diameter, layer span, and maximum aspect ratio:

Via type	Drill diameter	Layer span	Max aspect ratio
Microvia	≤6 mil	1–2 layers	<1:1
Stacked	≤6 mil	≥2–3 microvias	<1:1
Staggered	≤6 mil	Multiple via per stack	<1:1 each via

2.1.1 Difference between stacked and staggered vias

Features	Staggered vias	Stacked vias
Layout footprint	Requires more XY space due to offsets	Compact as it saves XY area
Routing density	Moderate	Very high; ideal under dense BGAs
Manufacturing	Simpler, as no copper fill is needed, and has fewer process steps	Complex as it needs copper fill and planarization
Reliability	High, especially during thermal cycling	Moderate as the risk of interface stress is high for more than 2 stacks
Signal integrity	Good as it provides short via paths	Excellent, since it creates a smooth vertical transition with controlled impedance
Cost	Lower than stacked vias	Higher due to more fab cycles and inspection

2.1.2 Manufacturing considerations for stacked and staggered vias



Cross-sectional view of stacked vias

1. Drilling

- Laser drilling produces tapered (frustum-shaped) holes. Hence, achieving precision is critical.
- Staggered vias are drilled individually for each layer, offering more tolerance.
- Stacked vias require highly accurate vertical alignment across layers, increasing complexity.

2. Sequential lamination

- Build-up layers are added one at a time (e.g., 1+n+1, 2+n+2), enabling controlled microvia creation.
- IPC-2226 outlines suitable stack-up configurations for both staggered and stacked via designs.

3. Copper fill and planarization

- Stacked vias generally require copper filling to create a solid, conductive via barrel, followed by planarization to maintain flatness.
- Staggered vias are typically left unfilled, reducing process steps, cost, and cycle time.

4. Inspection

- Advanced inspection methods like X-ray imaging and cross-sectioning are necessary to detect voids, misalignment, or delamination, especially in stacked via structures.

2.1.3 Design guidelines for reliable via structures

1. Limit stacked via depth to two layers

- Interconnect stress test studies show increased failure rates if more than 2 microvias are stacked.

2. Use a hybrid via strategy for HDI designs with three or more layers

- Combine stacked and staggered microvias; use stacked vias only where high routing density demands it.

3. Maintain adequate via-to-via spacing

- Ensure center-to-center spacing is at least equal to the drill diameter (e.g., ≥6 mil) to reduce mechanical and thermal stress.

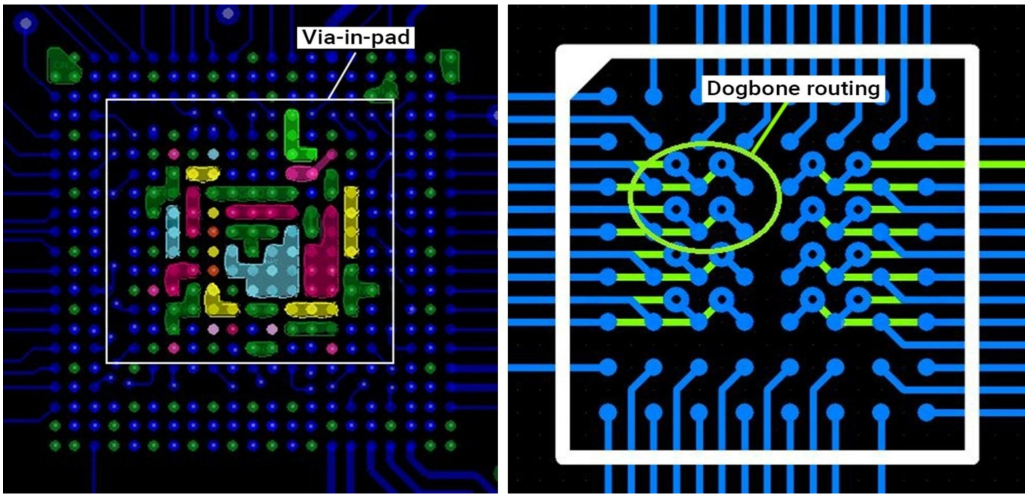
4. Keep the microvia aspect ratio below 1:1

- Maintain aspect ratio below 1:1 to ensure reliable plating and reduce the risk of via failure during thermal or mechanical stress.

5. Consult your fabricator during the design phase

- Staggered vias may not require copper fill or planarization, which can reduce manufacturing costs. Always confirm this with your fab house.

2.2. Via-in-pad



Via-in-pad and dog bone routing techniques

In via-in-pad (VIP) designs, vias are placed directly within component pads, commonly under BGAs or dense surface-mount devices (SMDs). These vias connect the pad to internal or bottom layers, reducing routing congestion.

This approach improves space utilization, thermal performance, and electrical characteristics, but it also adds manufacturing complexity. Hence, careful design and close collaboration with the fabricator are essential.

IPC-4761 type VII vias, which are filled and capped with plated copper, are best suited for via-in-pad structures in HDI PCBs. They provide a flat, reliable surface for component mounting and improve overall structural integrity.

2.2.1 Key benefits of via-in-pad

1. Improved form factor: Saves valuable XY area by placing vias directly under component pads.

2. Improved routing density: Eliminates the need for dog-bone fan-out, simplifying trace routing in dense areas.

3. Lower parasitics: Reduces trace length and inductance, ideal for high-speed signal paths.

4. Better thermal conduction: When filled with conductive material, vias act as effective thermal pathways.

5. Enhanced signal integrity and EMI performance: Shorter return paths help minimize noise in high-speed and RF circuits.

2.2.2 Typical applications of VIP

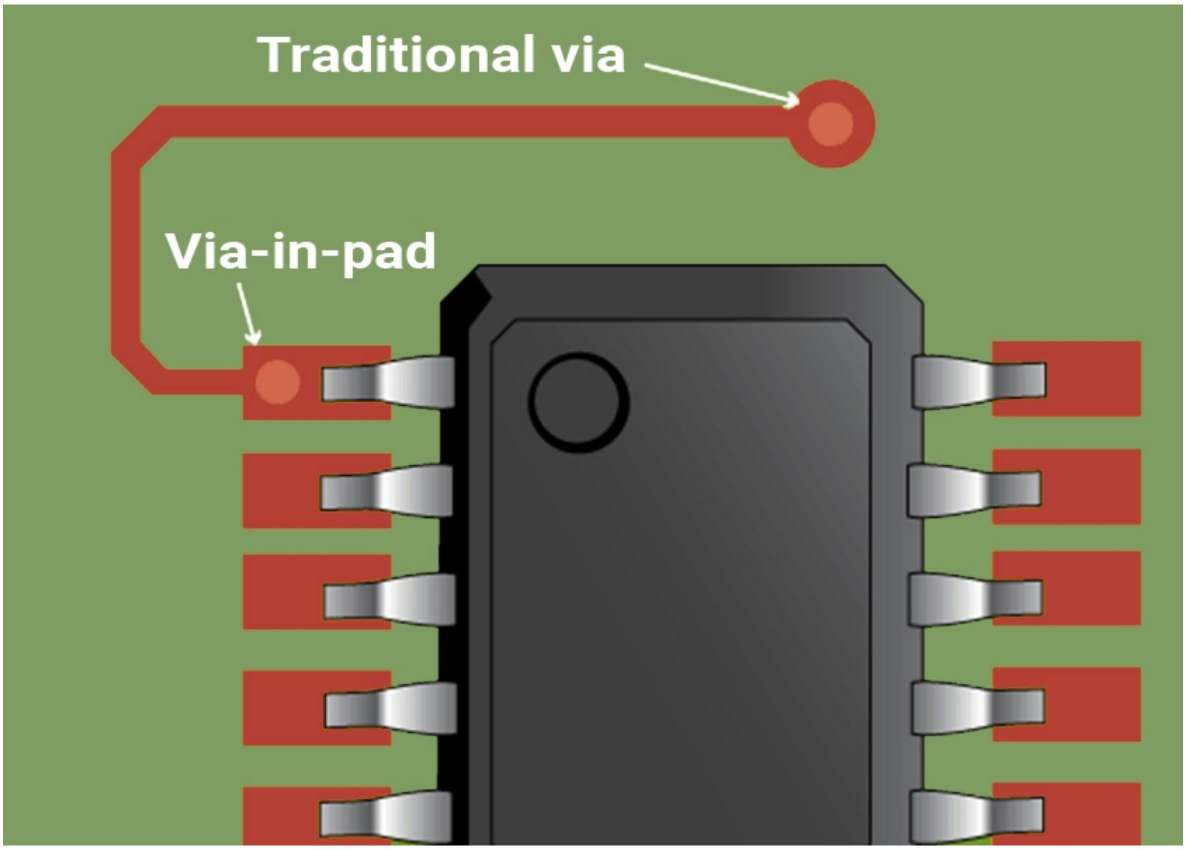
1. Fine-pitch BGAs (≤0.8 mm): Enables clean escape routing where traditional methods aren’t feasible.

2. High-speed digital lines: Reduce stub length and inductive effects for improved signal quality.

3. Power and thermal pads: Help dissipate heat efficiently from high-power components.

4. RF and EMI-sensitive areas: Supports controlled impedance and effective shielding in critical signal regions.

2.2.3 Manufacturing considerations for via-in-pad designs



An illustration showing a via-in-pad and a traditional via

1. Via filling

- Vias must be filled with conductive or non-conductive epoxy to prevent avoids. This stops solder from flowing into the via during assembly, which can cause poor solder joints or tombstoning.

2. Capping and planarization

- Once filled, vias are capped with copper and planarized to create a smooth, level pad. This is critical for proper component placement and reliable soldering.

3. Fill materials with matched CTE

- Filled vias may expand differently than the surrounding PCB materials. To prevent cracks during reflow, select fill materials with a coefficient of thermal expansion (CTE) that matches the laminate.

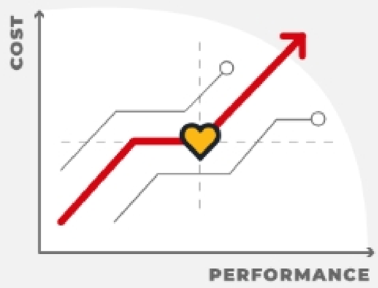
4. Drilling

- Whether laser- or mechanically-drilled, vias must be precisely aligned under component pads. Misalignment or oversized vias can lead to connectivity issues or distorted pads.

5. IPC-compliant filling and capping

- Poorly filled or uncapped vias can cause solder wicking, open joints, or reflow defects. Always confirm that your fabricator follows IPC standards for via-in-pad processing.

Via-in-pad adds cost and increases lead time. Reserve it for critical applications such as fine pitch BGA fanout or for impedance-controlled nets.



PCB DESIGN TOOL

Material Selector

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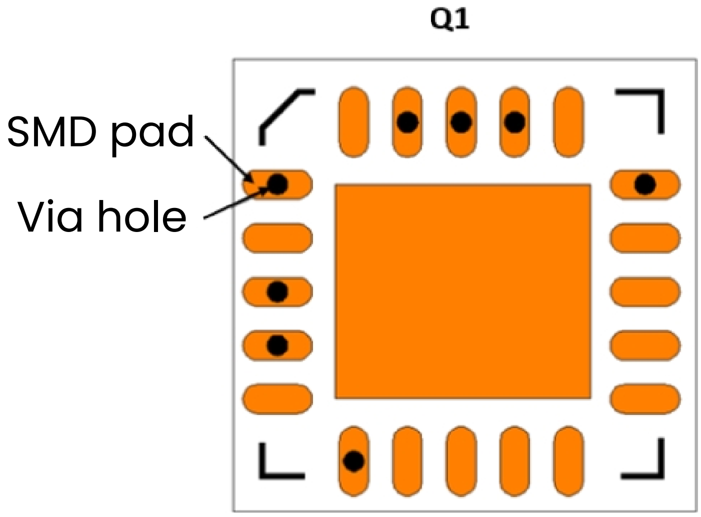
Best design practices for via-in-pad



Via-in-pad design for a QFN

1. Use IPC-4761 type VII vias

- These are fully filled and copper-capped, providing the flat, reliable surface required for via-in-pad designs.

2. Avoid open or tented vias-in-pad

- Unfilled or tented vias can lead to solder wicking, resulting in weak or incomplete solder joints.

3. Optimize via size for fine-pitch components

- Keep via diameters small (typically ≤ 7.87 mil) to fit within tight fanout areas in BGAs, CSPs, and QFNs.

4. Design pads with proper solder mask clearance

- Ensure pad dimensions allow for accurate stencil alignment and adequate solder mask relief.

5. Specify epoxy-filled and planarized vias

- Whether conductive or non-conductive, filled vias with planarized copper caps ensure a smooth, solderable pad surface.

6. Evaluate thermal and mechanical stresses

- In high-density areas, consider how stress may affect critical components, especially during reflow.

7. Maintain signal integrity for high-speed lines

- Use impedance-controlled routing and minimize stubs with techniques like back drilling or controlled-depth drilling.

8. Account for fabrication tolerances

- Precision is key in laser drilling, via filling, and surface planarization, design with tight tolerances in mind.

9. Use NSMD (non-solder mask defined) pads

- NSMD pads allow better solder fillet formation and are easier to inspect optically.

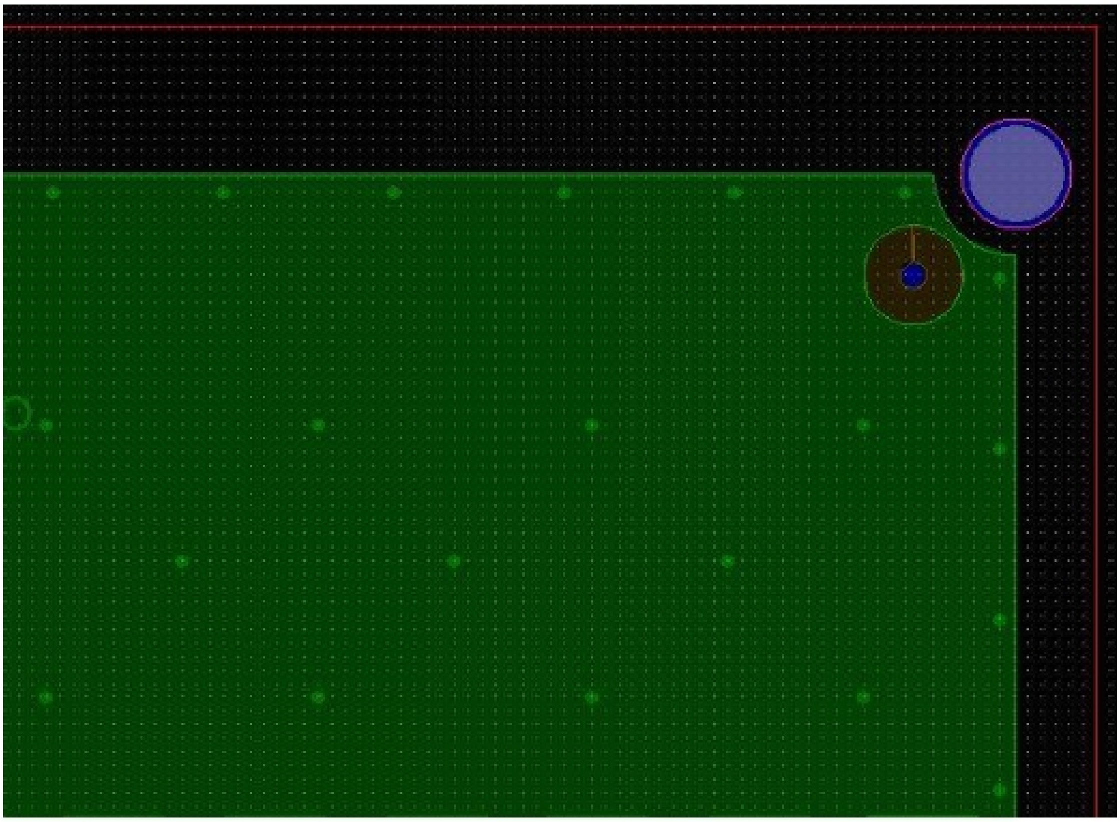
10. Avoid exposing via openings

- Do not leave mask openings over vias unless they are completely filled and plated.

11. Work closely with your fabricator to define:

- The via structure and stack-up strategy.
- Type of fill material (conductive or non-conductive).
- Surface finish compatibility (e.g., ENIG, OSP).

2.3 Via stitching



A PCB layout with via stitching (white dots)

Via stitching is an array of plated holes connecting copper pours or ground/power planes on different layers of a printed board.

2.3.1 Key benefits of via stitching

1. Low-impedance return path: Provides efficient return current paths, reducing voltage drop and noise.

2. Potential equalization: Helps maintain a uniform ground potential across all planes.

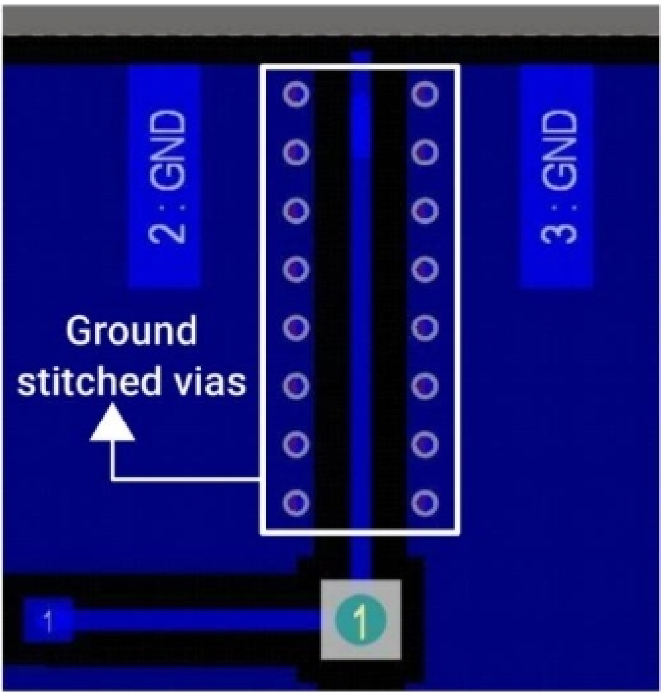
3. Reduced loop area and crosstalk: Minimizes EMI and improves signal quality in high-speed or differential pair routing.

4. Analog/RF isolation: Segregates sensitive analog or RF zones from noisy digital areas.

5. Improved thermal conduction: Enhances vertical heat flow into internal layers for better thermal management.

6. Better plane coupling: Lowers both DC and AC impedance between planes, supporting signal and power integrity.

2.3.2 Design guidelines for ground stitching vias



A PCB layout with ground stitching vias

1. Distribute stitching vias systematically across ground planes

- Maintain a minimum spacing of ~100–500 mil to minimize ground impedance and ensure effective EMI shielding.

2. Use standard via dimensions for thermal and electrical reliability

- A typical via (10 mil drill, 20 mil pad, 1 mil plating) offers ~1.5 mΩ DC resistance and 180 °C/W thermal resistance.
- Example: To carry 20 A safely, use 10 stitching vias to limit current per via to 2 A and temperature rise to ~1 °C.

3. Perform stitching after the copper pours are finalized

- This prevents placement conflicts and ensures maximum ground coverage.

4. Avoid placing stitching vias in sensitive analog zones

- Only include them if directly connected to the analog ground plane to prevent noise coupling.

5. Ensure proper antipad clearance

- This maintains plane isolation. For floating copper areas, stitching vias can be added post layout to improve connectivity.

2.4 Via fencing (shielding)

Via fencing involves placing one or more rows of grounded vias, often described as a "picket fence," alongside critical traces or board edges to suppress EMI and enhance signal integrity.

2.4.1 Advantages of incorporating via fencing in your PCB layout

1. Electromagnetic field containment: Minimizes EMI leakage and enhances signal integrity by surrounding critical traces.

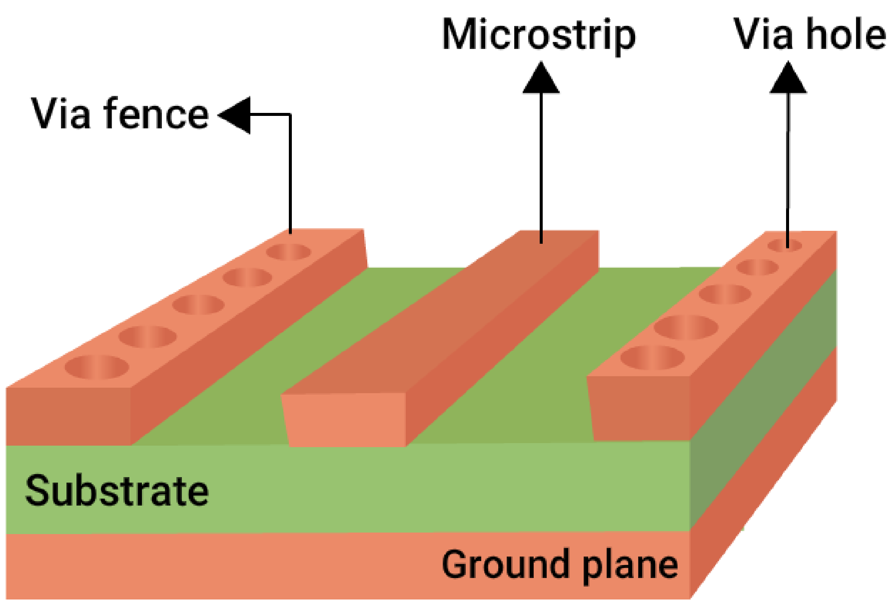
2. High-frequency shielding: Acts as a via curtain to isolate RF or high-speed zones from surrounding circuitry.

3. Effective EMI suppression: Improves noise isolation in high-speed and high-frequency environments.

4. Reduced loop inductance: Lowers inductance at vias and traces, supporting better high-speed signal performance.

5. Analog signal protection: Shields sensitive analog paths from nearby digital switching noise, preserving signal fidelity.

2.4.2 Tips for designing effective via fencing



An illustration of a PCB with via fencing

1. Use tight via spacing for high-frequency shielding

- For RF applications, keep via spacing ≤ λ/20 at your highest frequency to ensure effective EMI containment.

- Example: At 10 GHz with εr >≈ 4, the maximum via pitch should be ~12 mil.

- Use this formula to calculate the via spacing:

$$\text{Spacing} \leq \frac{\lambda}{20} = \frac{c}{20 \times f \times \sqrt{\varepsilon r}} \quad (c\text{: speed of light})$$

- For non-RF regions, a via pitch of 50–100 mil (1.27–2.54 mm) is typically sufficient for general shielding.

2. Place vias close to critical traces

- Keep fencing vias as close as possible to high-speed or RF traces, within DRC limits, to reduce field leakage.

3. Maintain parallel and enclosed geometries

- Align fences parallel to trace edges. Use U-shaped or closed-loop fences around sensitive zones for better isolation.

4. Use 1–2 rows along the microstrip edges

- Reinforce shielding by placing one or two rows of grounded vias along microstrip or waveguide edges.

5. Ensure solid ground connectivity

- Connect all fencing vias to a continuous ground plane. Avoid isolated copper islands, every via must have a valid return path.

6. Stitch into outer or inner ground pours

- Connect fencing vias to solid ground pours on adjacent layers to enhance shielding effectiveness.

7. Maintain shielding across layer transitions

- Extend fencing vias vertically through all signal layer transitions to preserve EMI protection throughout the stackup.

8. Use multiple ground layers for return continuity

- Support consistent return paths by utilizing multiple ground layers beneath high-speed signals.

9. Follow clearance and DFM rules

- Ensure via-to-via and via-to-trace spacing complies with your fabricator's DFM guidelines.

- Avoid over-constraining routing by balancing via density with available layout space.

10. Use antipads when vias cross signal layers

- Fencing vias should have proper antipads when they intersect signal layer to prevent short and parasitic coupling.

11. Leverage thermal and mechanical benefits

- Via fences also act as thermal vias in high-power zones, helping dissipate heat into internal layers.

- They add mechanical strength near board edges or cutouts, reducing flex and improving reliability.

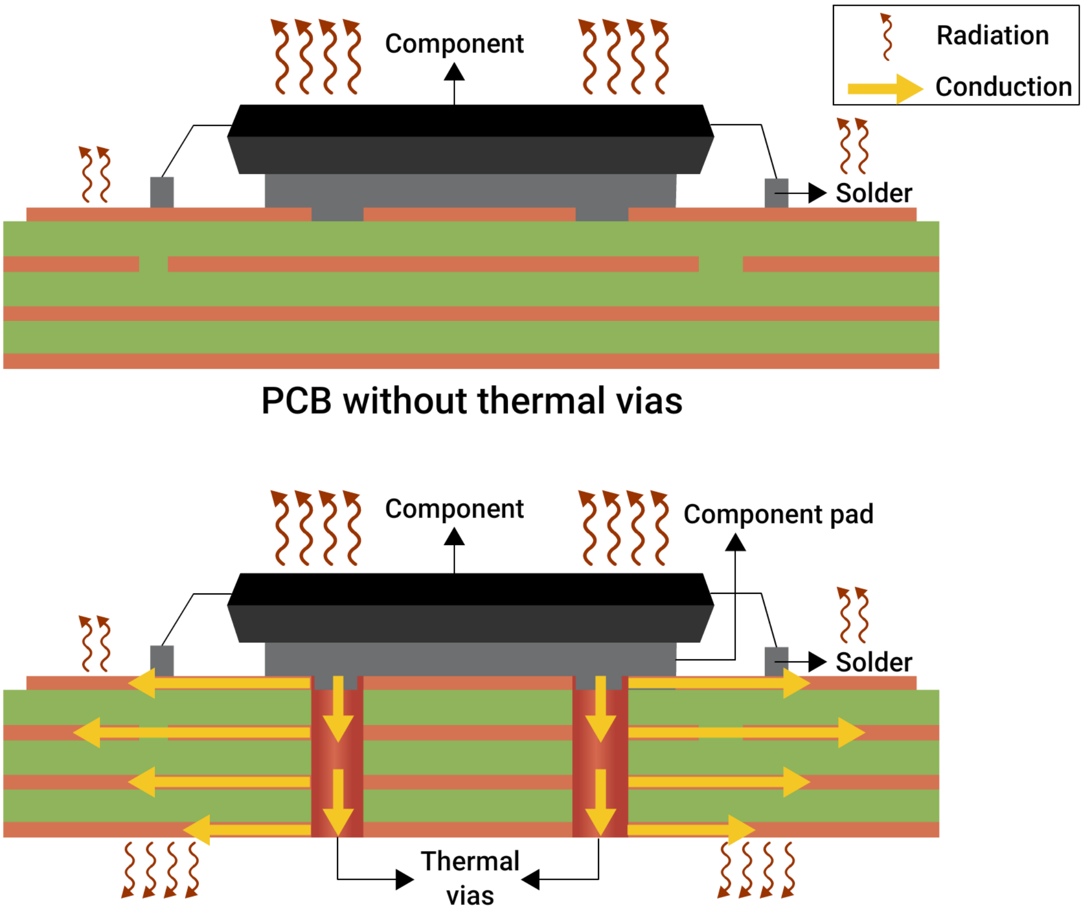
12. Take advantage of EDA tool automation

- Use fencing/stitching features in tools like Altium Designer, Proteus, Zuken, or Cadence Allegro to automate layout and save time.

2.4.3 Via stitching vs. via fencing

Features	Via stitching	Via fencing
Purpose	Connects ground planes and helps distribute power and heat	Isolates signals or zones to block EMI and reduce interference
Pattern	Dense arrays spread across the board	One or more via rows placed along critical signal paths
Spacing	100–500 mil between vias	Tight spacing ≤ λ/20 at the highest frequency (e.g., ~0.3 mm (12 mil) at 10 GHz)
Layers connected	Connects all signal, power, and ground planes	Usually placed on the outer layer and tied to a ground plane
Benefits	Lowers return path impedance and improves thermal performance	Blocks EMI and minimizes crosstalk near sensitive signals

2.5 Thermal vias



Illustrations of PCBs with and without thermal vias

Thermal vias are an essential tool for transferring heat away from high-power components, ensuring optimal performance and prolonging product lifespan.

They are plated-through holes, typically unconnected to signals, that conduct heat from an external layer (often the top, where the component is mounted) to one or more internal or bottom copper planes. These are usually located under or near heat-generating components such as power ICs, MOSFETs, LEDs, and CPUs.

2.5.1 How do thermal vias work?

Copper is an excellent thermal conductor (≈ 400 W/m·K), and a plated via acts as a vertical thermal pipe. Thermal vias transfer heat vertically from a component's thermal pad to a large copper area, often on the internal or bottom layers, which then disperses the heat laterally or to a heat sink.

2.5.2 Thermal via design parameters

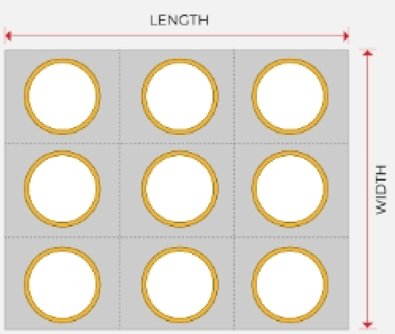
Parameters	Typical values/guidelines
Via drill size	6–16 mil
Via pad size	14–30 mil
Plating thickness	≥1 oz copper (≈ 1.4 mils or 35 µm)
Via pitch	20-60 mil (depends on component size)
Via fill type	Optional (non-conductive or conductive epoxy), depending on application
Connection	Always connect to large copper planes for effective heat dissipation

Thermal resistance of vias

The thermal resistance of a via depends on drill size, plating thickness, and how well it connects to surrounding copper planes. The table below shows approximate values for a standard configuration:

Via size and configuration	Thermal resistance (°C/W)
10 mil drill, 1 oz plating	~180 °C/W
10 parallel vias (same dimensions mentioned above)	~18 °C/W (in ideal contact)

Rule of thumb: One 10 mil via removes ~100–200 mW of heat with a 10–20°C rise. For higher power components (>1 W), you need multiple vias.



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Via Thermal Resistance Calculator

TRY TOOL

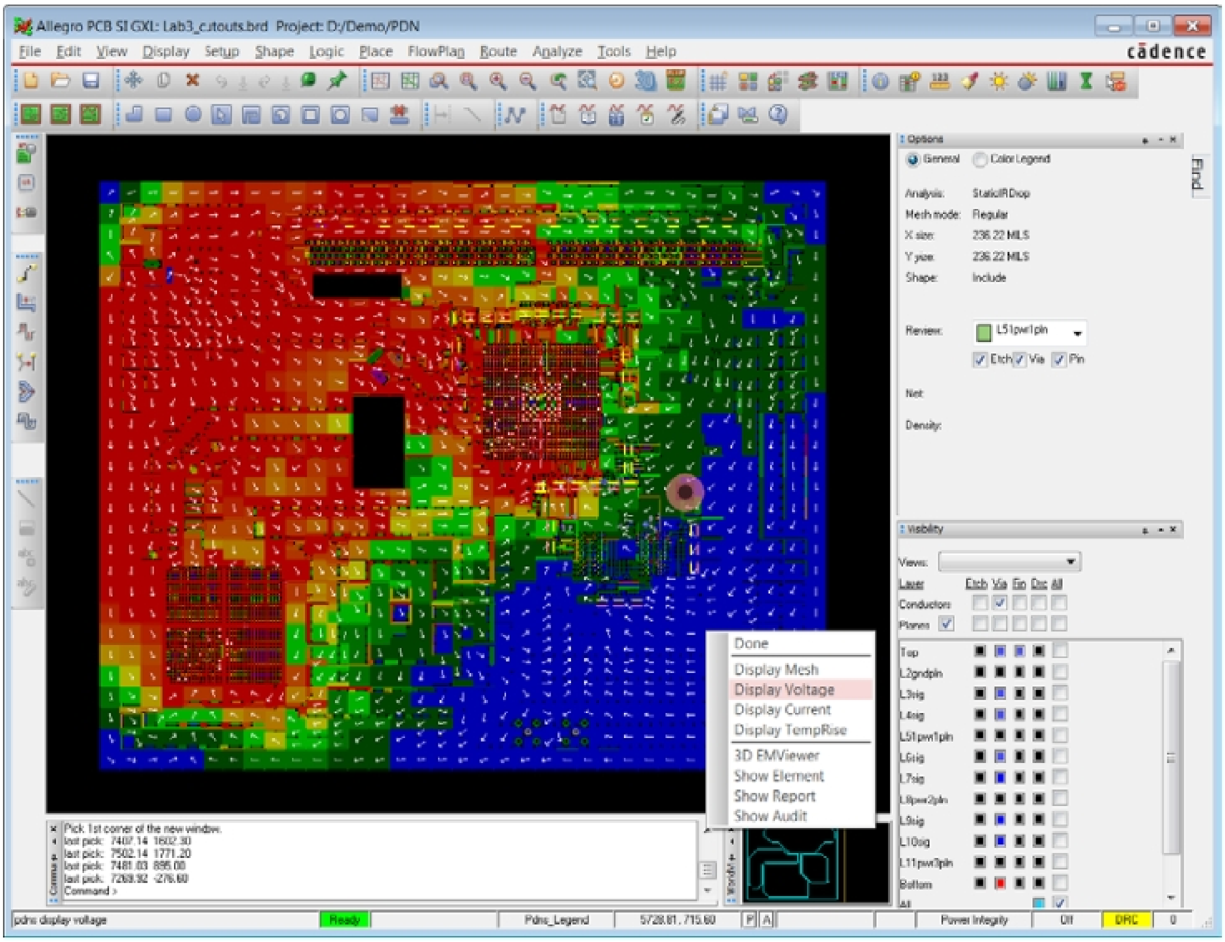
According to IPC-2221B and IPC-4761, thermal vias must meet standard mechanical and electrical reliability requirements. The following table summarizes key IPC-recommended parameters for reliable thermal via implementation:

Parameters	IPC recommendations
Aspect Ratio	≤10:1
Annular Ring	≥4 mil
Pad-to-plane clearance (antipad)	≥10 mil
Via fill (type VII for VIP)	Filled and capped (for thermal and assembly reliability)

To effectively manage heat dissipation, the number and arrangement of thermal vias should be scaled to match the power output of the component. The table below provides typical via count recommendations for common surface-mount power components:

Components	Power dissipation	Thermal via matrix
QFN-32 (3×3 mm)	1 W	4×4 thermal vias connecting to the bottom copper plane
Buck Converter (LM76003)	2 W	3×3 via array to a large bottom-side copper pour
LED Array (COB)	3–5 W	Dense via matrix to copper slug or external heatsink
MOSFET (D2PAK)	4–8 W	Filled vias connecting to a heat spreader zone or heatsink

2.5.3 Thermal simulation and analysis



Thermal simulation in Allegro

Thermal analysis helps validate whether your thermal via design can effectively manage heat under real-world operating conditions.

Use simulation tools like ANSYS Icepak, SolidWorks Flow Simulation, or Altium's PDN Analyzer to model heat transfer through thermal vias and copper planes.

2.5.4 Thermal via layout guidelines for efficient heat dissipation

1. Use large copper pours

- Increase copper area on inner layers to improve lateral heat spreading.

2. Maximize via density

- Place as many thermal vias as possible under heat sources, staying within spacing and clearance limits.

3. Connect vias to bottom layers

- For double-sided boards, connect thermal vias to bottom-layer copper or directly to a heatsink using thermal reliefs if needed.

4. Ensure good TIM contact

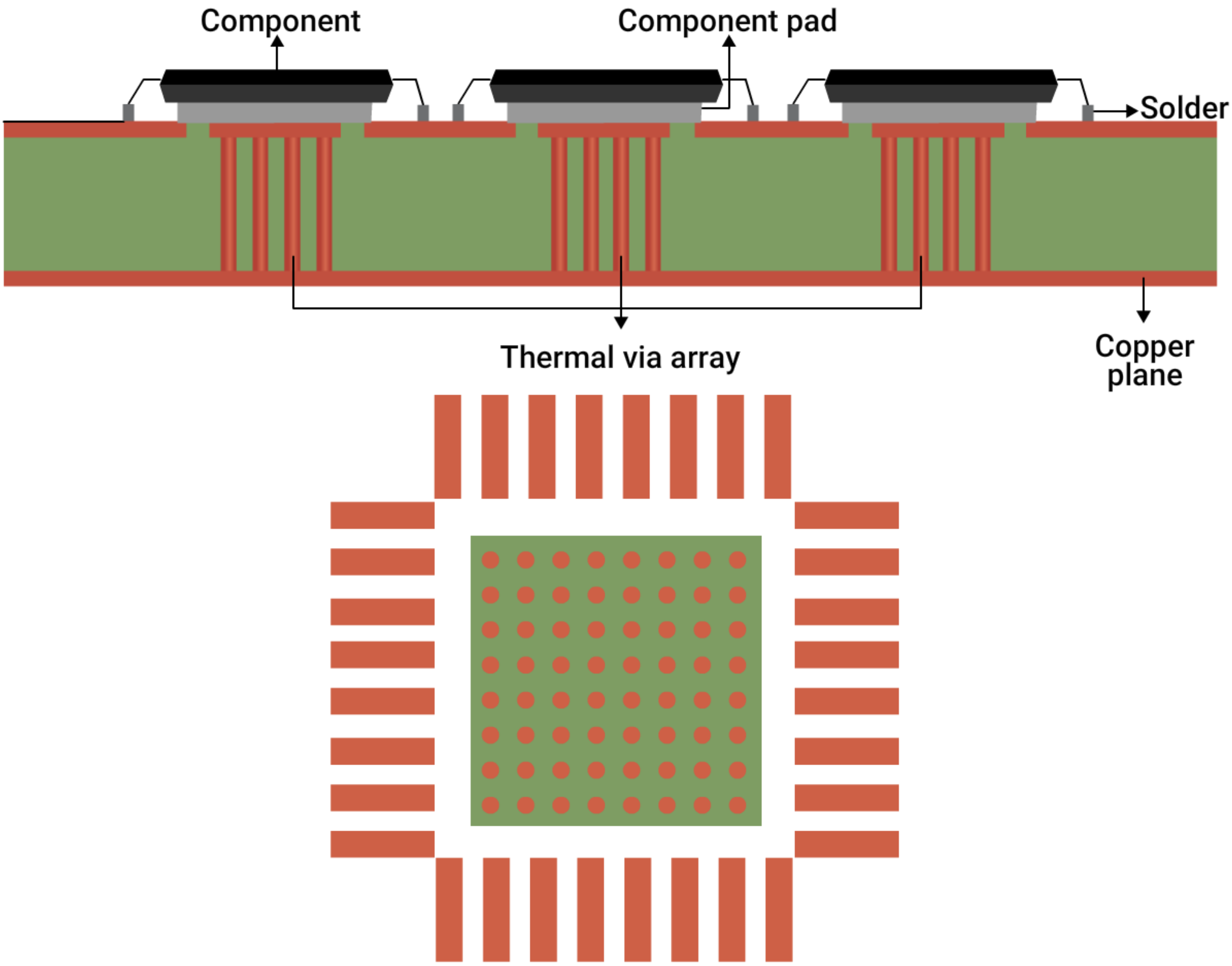
- When using thermal interface material (TIM), planarize via caps to ensure optimal contact with the heatsink.

5. Place vias beneath thermal pads

- Have thermal vias directly beneath the component's thermal pad (e.g., PowerPAD or exposed pad of a QFN).

6. Use via arrays

- Arrange vias in a matrix (e.g., 3×3, 4×4) to evenly distribute heat across the copper plane.



An illustration of a PCB with thermal vias

The above example features a via pitch of 1.2 mm and a diameter of 0.3 mm. This array is designed to efficiently transfer heat away from the component while remaining compatible with standard PCB manufacturing processes. The chosen spacing allows for reliable copper plating and minimizes the risks of via cracking and pad lifting.

7. Control solder paste flow

- Tented or plugged vias help prevent solder wicking during reflow, improving joint reliability.

8. Implement IPC type VII via for VIP

- For via-in-pad designs under thermal pads, use filled and capped vias as per IPC-4761 to maintain solderability.

9. Connect thermal vias to copper pours

- Always link thermal vias to internal or bottom-side copper planes or heatsinks for effective heat transfer.

10. Maintain optimum trace clearance

- Keep sufficient spacing from signal traces or sensitive analog areas to prevent thermal and electrical interference.

3. Common DFM mistakes in via design

Design for manufacturing (DFM) ensures that your PCB layout meets performance targets and is production-ready.

This section highlights frequently overlooked issues in via design, including incomplete or inconsistent drill data, missing via specifications, and improper via-in-pad definitions that can lead to manufacturing defects.

3.1 Drill data discrepancies

An example of a Gerber file with a drill map

Fabricators rely on accurate drill tables to determine drill sizes, drill registration, and which layers each vias connect. Any mismatch can lead to stack-up errors, drill breakage, misregistration, scrapped panels, and board respins.

Common drill data issues include:

- Mismatches between Gerber and NC drill files.
- Conflicting drill sizes assigned to the same net or via type.
- Missing layer span definitions for blind or buried vias.
- Drills not called out in fabrication notes.
- Mixed units (e.g., mm and mil) or missing unit declarations in Excellon files.

The table below provides the drill file documentation guidelines:

Things to include	Guidelines
Units	Clearly define in the header (inch or metric)
Drill sizes	Use standard sizes (e.g., 8, 10, 12 mil) unless required
Format	Use Excellon 2 format with leading zero suppression
Layer span info	Clearly specify. For instance, L1–L2 blind, L3–L6 buried
Drill symbols in Gerber	Match with the legend or drill chart
Drill map (optional)	Provide it for visual confirmation with net/via labels

Finally, ensure that each drill pair is explicitly documented in both the fabrication drawing and stack-up diagram. This is critical for HDI boards with multiple lamination cycles.

3.2 Missing drill files and specifications

A PCB cannot be fabricated without complete drill data. Missing or incomplete drill files are one of the most common DFM red flags, leading to production holds, delays, and multiple back-and-forths with your fabricator.

Even when drill files are present, missing key specifications such as plating thickness or hole tolerance can severely impact via reliability and board yield.

DFM reviews often reveal missing or incomplete drill data, such as:

1. Forgotten drill files during CAM file generation.

2. Missing drill legend or chart in Gerber or assembly documentation.

3. No tolerance values for hole sizes or annular rings.

4. Via types not defined (e.g., microvia, backdrilled, via-in-pad).

Symbol	Count	Hole Size	Plated	Hole Type	Hole Tolerance	Via/Pad	Drill Layer Pair
⌘	57	6.00mil	PTH	Round	+/-3.00mil	Via	Top Layer - Bottom Layer
▽	8	18.00mil	PTH	Round	+/-3.00mil	Pad	Top Layer - Bottom Layer
□	3	50.00mil	PTH	Rectangle	+/-3.00mil	Pad	Top Layer - Bottom Layer
○	1	39.37mil	NPTH	Round	+/-2.00mil	Pad	Top Layer - Bottom Layer
	69 Total						

Slot definitions : Routed Path Length = Calculated from tool start centre position to tool end centre position.
Hole Length = Routed Path Length + Tool Size = Slot length as defined in the PCB layout

Symbol	Count	Hole Size	Plated	Hole Type	Hole Tolerance	Drill Layer Pair	Via/Pad
⌘	13	6.00mil	PTH	Round	+/-3.00mil	Int1 - Int4	Via
	13 Total						

Symbol	Count	Hole Size	Plated	Hole Type	Hole Tolerance	Drill Layer Pair	Via/Pad
☼	9	4.00mil	PTH	Round	+/-3.00mil	Int1 - Int2 (GND)	Via
	9 Total						

Symbol	Count	Hole Size	Plated	Hole Type	Hole Tolerance	Drill Layer Pair	Via/Pad
☼	26	4.00mil	PTH	Round	+/-3.00mil	Top Layer - Int1	Via
	26 Total						

Symbol	Count	Hole Size	Plated	Hole Type	Hole Tolerance	Drill Layer Pair	Via/Pad
☼	6	4.00mil	PTH	Round	+/-3.00mil	Int3 (PWR) - Int4	Via
	6 Total						

Symbol	Count	Hole Size	Plated	Hole Type	Hole Tolerance	Drill Layer Pair	Via/Pad
☼	33	4.00mil	PTH	Round	+/-3.00mil	Int4 - Bottom Layer	Via
	33 Total						

An example of a drill chart

Must have specifications in your drill files

As fabricators, we have observed that designers generally miss out on the following data:

Parameters	Recommendations
Hole type	Clearly define: PTH, NPTH, microvia, VIP, backdrilled
Tolerance (plated holes)	±3 mil
Tolerance (NPTH)	±2 mil
Aspect ratio	≤10:1 (mechanical), ≤1:1 (laser-drilled microvia)
Annular ring	≥4 mil (Class 2), ≥5 mil (Class 3)
Plating thickness	≥0.8 mil (20 µm) barrel (IPC-6012)
Drill chart	Include drill size, count, and symbol

*These values can vary depending on the PCB fabricator's capabilities.

Drill file submission checklist:

- NC Drill file (.drl or .txt).
- Drill tool list (.tol).
- Drill legend/chart in assembly drawing.
- Layer span information (especially for HDI).
- Stacked/staggered microvia construction notes.
- Backdrill depth and clearance callouts.

3.3. Incorrect via-in-pad details

Via-in-pad (VIP) is a common technique used to route signals under dense packages like QFNs, BGAs, and CSPs. However, it requires special fabrication processes, including:

- Via filling (conductive or non-conductive epoxy).
- Planarization and copper capping.
- Paste mask adjustments.

If these vias are left unfilled, solder wicks into the hole, creating voids and cold joints, especially under QFNs, where joints are not visible for inspection. Hence, specifying clear details about the via-in-pad in production documentation becomes crucial.

Commonly observed issues with the via-in-pad specification include:

- VIPs are shown in Gerbers but not mentioned in fabrication notes.
- Fill type (conductive/non-conductive) not specified.
- No information on planarization or copper capping.
- Solder mask openings left on unfilled vias.
- VIPs are not clearly flagged in the drill chart or netlist.

Best practices for documenting via-in-pad structures

1. Define via structure using IPC-4761 classifications

- E.g., Type VII (filled and capped with plated copper) is recommended for VIPs under fine-pitch BGAs.
- Include in the fabrication notes:
    1. Via type (e.g., microvia, through-hole).
    2. Construction (e.g., filled with non-conductive epoxy and copper capped).
    3. Purpose (e.g., BGA escape routing, thermal relief).

Fab note sample: `Use IPC-4761 Type VII via-in-pad structure for all VIPs beneath fine-pitch BGAs. Fill with non-conductive epoxy and cap with plated copper.`

2. Provide complete stack-up and layer pair information

- Include a detailed stack-up drawing showing VIP locations and layer spans (e.g., L1–L2, L2–L3), especially for buried or staggered VIPs.

3. Specify via filling and capping parameters:

- Fill material: conductive or non-conductive epoxy.
- Capping: Specify plating and planarization requirement.
- Void tolerance: specify whether partial fills are allowed.

Fab note sample: `All via-in-pad via must be 100% filled with non-conductive epoxy and copper capped to ensure a flat surface.`

4. Identify surface finish compatibility

- Certain finishes, such as ENIG and OSP, impact solder joint quality when used with VIPs. Always confirm this with your manufacturer.

Fab note sample: `For all VIPs under BGA, use ENIG surface finish for better planarity and solder joint quality.`

5. Address the solder mask and pad design

- Specify NSMD or SMD pad types (NSMD usually preferred).
- Confirm no solder mask opening over VIPs.
- VIP barrels, unless filled and capped.
- Define mask clearance around VIPs.

6. Specify key assembly considerations

- Flatness requirement over VIPs (e.g., ±15 µm).
- Stencil design guidelines to prevent excess solder paste.

Fab note sample: `VIPs must be flush with the pad surface with a maximum 15 µm step height for stencil compatibility.`

7. Make sure VIPs are consistently defined across drill files (Excellon), ODB++/IPC-2581/Gerbers.

- Label VIPs in drill tables or netlists (e.g., VIP_BGA_U1) for clear identification.

3.4. Drill design issues

Improper drill planning can compromise PCB manufacturability. Common design issues related to drills include:

- Drill sizes exceeding pad sizes.
- Inconsistent or missing mounting hole references.
- Use of unavailable or non-standard drill bit sizes.
- Not specifying plated and non-plated holes.
- Overlapping or duplicate drill locations.
- Inclusion of non-functional drills without purpose.

3.4.1 Best practices to prevent drill design errors

1. Maintain proper annular ring

- Ensure pad diameters are sufficiently larger than drill sizes to preserve annular ring integrity and enable reliable plating.

2. Provide complete drill data

- Include well-formatted Excellon drill files (.drl, .tol) in the fabrication package.

3. Use standard tool sizes

- Avoid very large or uncommon drill diameters (e.g., >400 mil) unless confirmed with the fabricator; consider routing alternatives if needed.

4. Clearly define plating status

- Specify whether each hole is plated (PTH) or non-plated (NPTH) in both the layout and drill documentation—especially for holes contacting copper.

5. Eliminate duplicate/overlapping drills

- Run thorough DRC and CAM checks to catch and remove overlapping or conflicting hole entries.

6. Properly classify non-functional holes

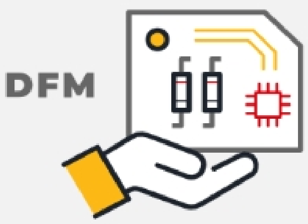
- Treat holes with no electrical connection on any layer as NPTH mechanical holes unless otherwise required.

7. Avoid redundant drill sizes at the same location

- Prevent stacked drills with different diameters to avoid confusion.

8. Stick to industry-standard sizes and formats

- This ensures compatibility with standard tooling, reduces lead times, and improves overall fabrication efficiency.



PCB DESIGN TOOL

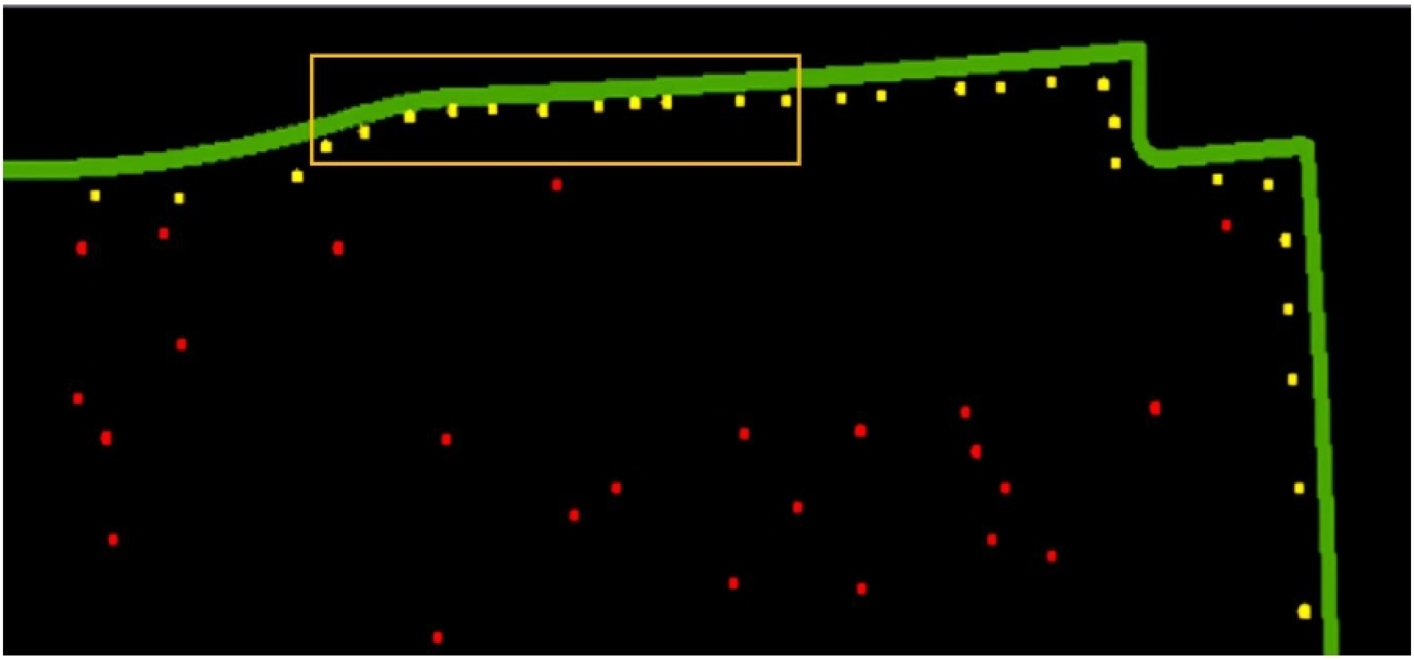
Better DFM

TRY TOOL

3.5 Inadequate drill clearances

Clearance violations are among the most common causes of shorts, fabrication defects, and test failures. Common issues include:

- Insufficient drill-to-copper clearance.
- Drill-to-drill spacing violations.
- Overlapping antipads on dense internal layers.
- Holes placed too close to the board edge.



Insufficient drill-to-board edge clearance

3.5.1 What every PCB designer should know about drill spacing

1. If vias or holes are placed too close to traces, pads, or copper planes, they can cause etch bridging, arcing, impedance distortion, or even mechanical breakout. Below are recommended minimum clearances to avoid such issues:

Feature type	Recommended clearance
Via-to-trace	6–8 mil
Via-to-SMD pad	10 mil
Drill-to-plane (antipad)	≥ 12–20 mil (based on voltage)

2. Antipads are clearance holes in internal plane layers that isolate vias from copper planes. Missing or incorrect antipads can result in via shorting, especially in multilayer designs.

3. Use net-driven rules in your EDA tool or automate antipad generation during CAM processing to ensure consistency.

4. In HDI designs, closely packed vias may lead to overlapping antipads, reducing net isolation, and causing leakage currents or EMI. Stick to these drill-to-drill spacing standards.

Drill type	Minimum drill-to-drill spacing
Mechanical	8–10 mil
Laser	≥5 mil

5. Drills near the PCB edge can lead to cracking, breakout, or panelization problems during fabrication or depaneling. Keep drilled holes at least 10–20 mil away from the board outline.

6. Always check your fabricator’s capabilities. Every PCB manufacturer has their own minimum capability specifications. If your drill clearances fall below these thresholds, the board might be rejected or require more expensive processing.

Key parameters to confirm with your fabricator:

1. Minimum drill size.

2. Minimum annular ring size.

3. Minimum trace/space.

4. Minimum drill-to-copper feature clearance.

4. Crafting perfect vias for your PCBs

Designing vias that are both functional and manufacturable requires a detailed understanding of the fabrication process and its constraints.

In this section, you'll find practical tips on critical manufacturing factors such as drill and pad sizes, aspect ratio limits, and smart via placement techniques to minimize the number of sequential lamination cycles.

4.1. Manufacturing considerations when designing a PCB via

Designing a via involves more than just picking a hole size in your layout tool. Each via must strike the right balance between electrical performance, manufacturability, and cost.

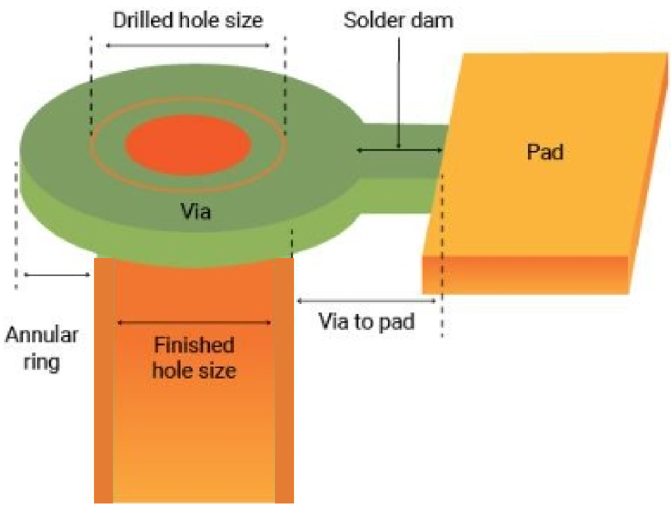
Key factors like choosing the appropriate drill and pad dimensions, ensuring reliable plating, and staying within your fabricator's limits are all critical to a successful via design.

4.1.1 Calculating drill, pad size, antipad, and annular ring sizes

Precise via geometry is critical for ensuring mechanical strength and signal integrity. The key elements to consider are:

Parameters	Description
Drill size	Diameter of the hole before plating.
Pad size	Copper area surrounding the hole.
Antipad	Clearance between the via pad and surrounding copper planes (on inner layers).
Annular ring	Distance from the edge of the drilled hole to the edge of the pad.

Drill size



Different drill parameters in a PCB design

Drill size is typically selected based on the via's current-carrying capacity. A larger drill supports thicker copper plating and offers a greater cross-sectional area for current flow.

Using an undersized drill for high-current applications can lead to excessive heat buildup, increasing the risk of delamination and reliability issues.

The following formula is used for calculating the via current capacity:

$$I = (K)(\Delta T^{\beta 1} / A^{\beta 2})$$

Where:

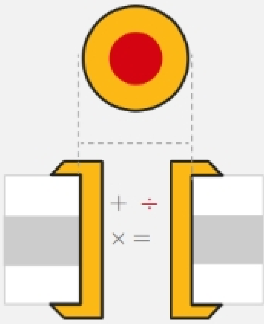
I	=	Current in amperes
ΔT	=	Temperature change with respect to ambient temperature in °C
A	=	Cross-sectional area in mils
K (correction factor)	=	0.024 (internal conductor) and 0.048 (external conductors)
β1	=	0.44
β2	=	0.75

Typical current ratings for standard vias

Via size (drill x plating)	Typical current capacity	Thermal resistance (°C/W)
10 mil, 1 oz plating	~1.0–1.5 A	~180 °C/W
12 mil, 1 oz plating	~2.0–2.5 A	~120 °C/W
16 mil, 1 oz plating	~3.0–4.0 A	~90 °C/W

Keep these pointers in mind when you're deciding on current carrying capacity:

- Implement multiple vias to reduce current per via and improve heat dissipation. For example, to safely carry 6 A, use at least 3–4 vias of 12 mils each.
- Use thermal via arrays under power devices to assist in heat sinking.
- Always consult your PCB manufacturer for exact via current rating charts, especially for high-power or automotive applications.

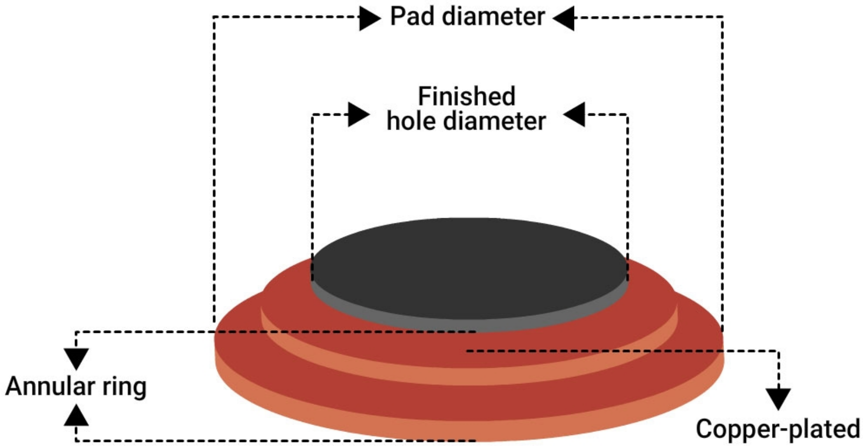


PCB DESIGN TOOL

Via Current Capacity and Temperature Rise Calculator

TRY TOOL

Pad size



Pad features

Pad design plays a crucial role in determining the structural and electrical reliability of vias. The pad must be large enough to accommodate proper annular ring width while also maintaining clearance for routing and solder mask.

Consider these guidelines when you're designing via pads:

1. Ensure the pad is well-centered over the drilled hole to avoid breakout.

2. Account for drill wander by adding tolerance in pad sizing.

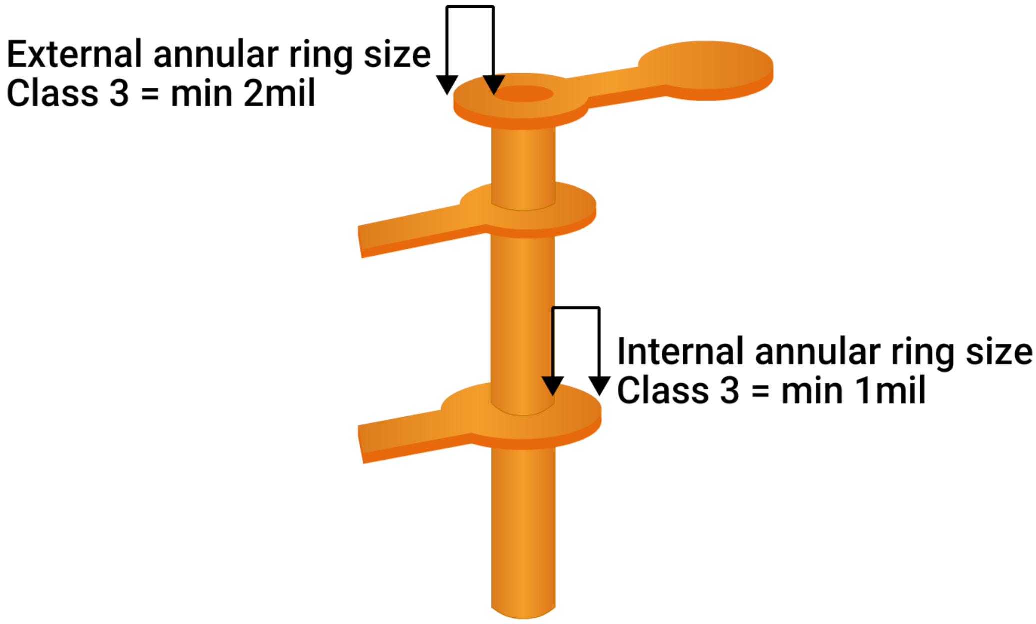
Minimum pad size = Drill size + (2 × Annular ring) + Fabrication tolerance.

3. Use non-solder mask defined pads where possible to improve solder joint quality.

4. Apply teardrop shapes at pad-to-trace junctions to reduce stress and prevent cracking.

5. Maintain minimum pad-to-pad and pad-to-trace clearances as per the fabricator's spacing rules to avoid shorts and improve yield.

Annular ring size



An illustration of internal and external annular rings

The annular ring ensures a strong electrical and mechanical connection between the via barrel and the copper pad. Its size impacts the via's resistance to thermal and mechanical stress.

Use this formula to calculate annular ring:

Annular ring = (pad diameter – drill diameter) / 2

Example:

For a 10-mil drill and 20-mil pad,

Annular ring = (20 – 10) / 2 = 5 mils

Best annular ring design practices:

1. For class 2 boards, have a minimum of 4 mil annular ring.

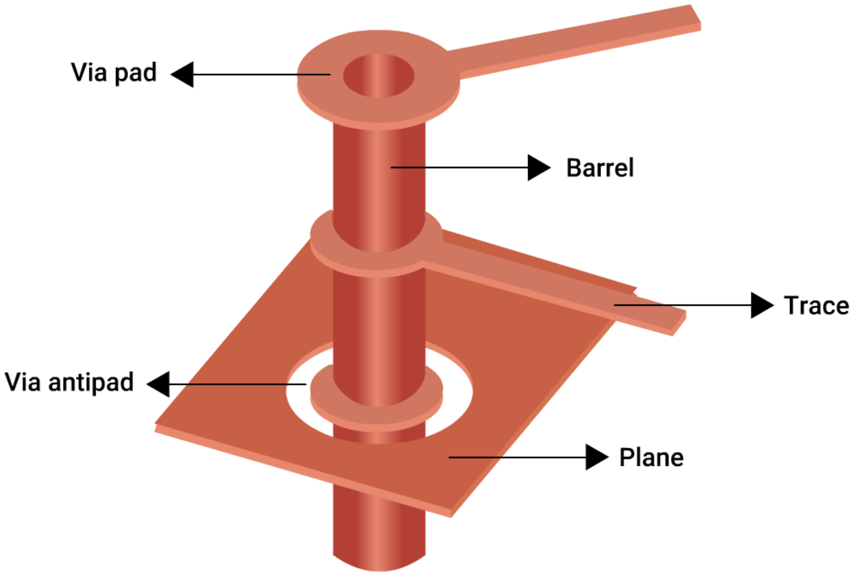
2. For class 3 PCBs, a minimum of 7 mil annular ring is required.

3. For microvias, 2-3 mil annular rings may be acceptable depending on aspect ratio and stack-up.

4. Ask your manufacturer to avoid tangency or breakout conditions; ensure copper remains completely around the hole.

Larger annular rings are easier to fabricate and inspect, while smaller rings (especially for HDI) require tighter registration control. Always coordinate with your fabricator to balance reliability and manufacturability.

Antipad guidelines



An illustration of an anitipad in an inner layer of a PCB

Antipads are voids or clearances in copper planes used to isolate vias from internal power or ground layers. These clearances are essential to prevent unwanted electrical connections and short circuits, especially in multilayer boards.

Design tips:

1. Use this rule of thumb for thermal reliefs:

   Antipad = Pad size + 10 mil

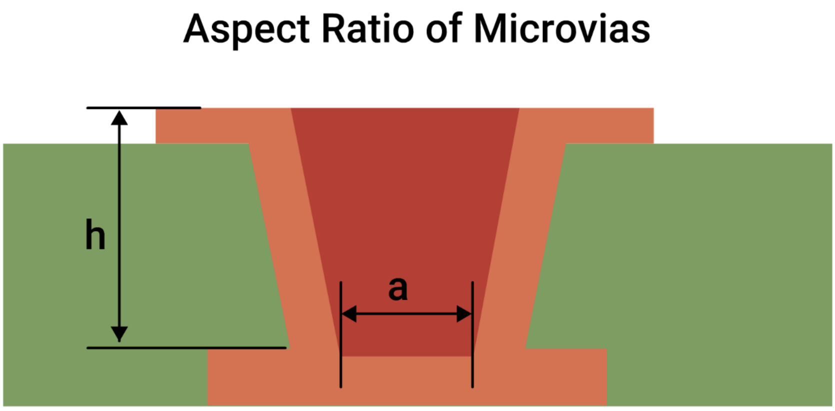
2. As per IPC-4761, follow these recommendations:

   Antipad = Drill size + 16 mil

   Plane relief = 8 mil

3. IPC-2221 specifies a minimum antipad clearance of 10 mil.

Aspect ratio



Microvia AR = h/a

Microvia aspect ratio calculation

The aspect ratio is the ratio between the PCB thickness and the finished drill diameter. It directly affects the quality of copper plating inside the via barrel.

A higher aspect ratio makes it harder for plating chemicals to reach and coat the hole uniformly. This can increase the risk of voids, poor connectivity, and cracking under thermal stress.

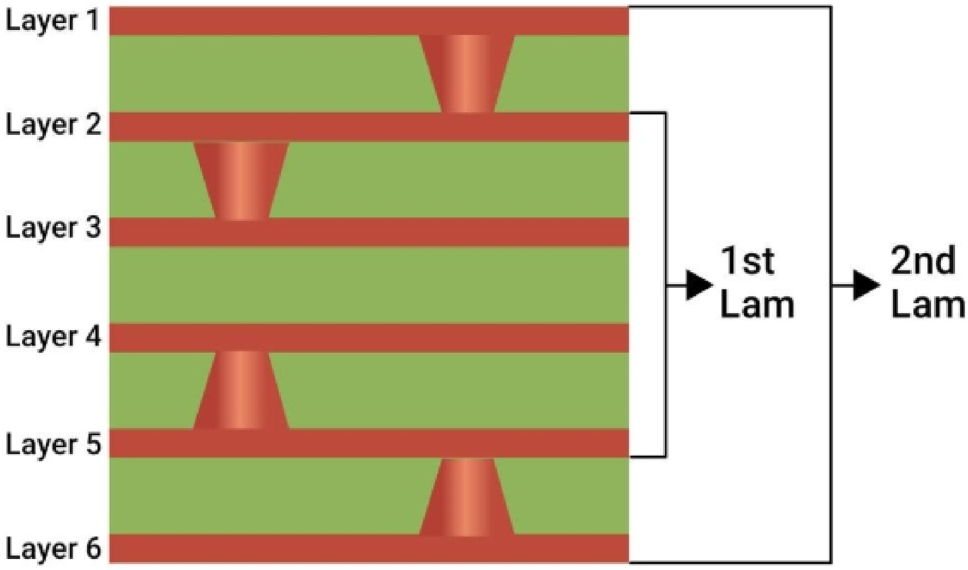
Aspect ratio = board thickness/drill diameter

Aspect ratio guidelines for different types of vias:

Via type	Maximum aspect ratios	Remarks
Through-hole	10:1	Reliable range with low risk
Blind via	1:1 to 0.75:1	Shallower vias for better plating
Microvia	0.75:1	Laser-drilled; max depth = 0.25 mm (typical)

Higher aspect ratios increase plating difficulty and the risk of avoids or cracks under thermal stress. Always work within your fabricator's capabilities.

4.1.2 Sequential lamination for blind, buried, and microvias



Sequential lamination process

Sequential lamination is a process of fabricating a circuit board using subsets composed of copper and dielectric layers. This process enables you to incorporate advanced via structures such as stacked and staggered microvias in your HDI PCBs.

Sequential lamination process:

1. Core fabrication: Drill and plate the core (e.g., L2-L3).

2. First lamination: Add prepreg and copper foil, then laser drill microvias (e.g., L1–L2).

3. Subsequent lamination cycles: Add additional layers (L4+, etc.) and drill/stitch more vias as needed.

HDI structure	Description
1-n-1	One microvia layer on each side of the core
2-n-2	Two sequential buildup layers per side
Staggered vias	Non-aligned microvias on different layers
Stacked vias	Vertically aligned microvias across layers

Follow these tips to reduce the lamination cycles:

1. Group vias with similar depth and connectivity to reduce unnecessary lamination cycles.

2. Avoid scattering blind or buried vias across unrelated layer pairs unless absolutely needed.

3. Use symmetric layer builds and mirror via structures where possible.

4. Minimize layer transitions that require stacked or nested blind/buried vias.

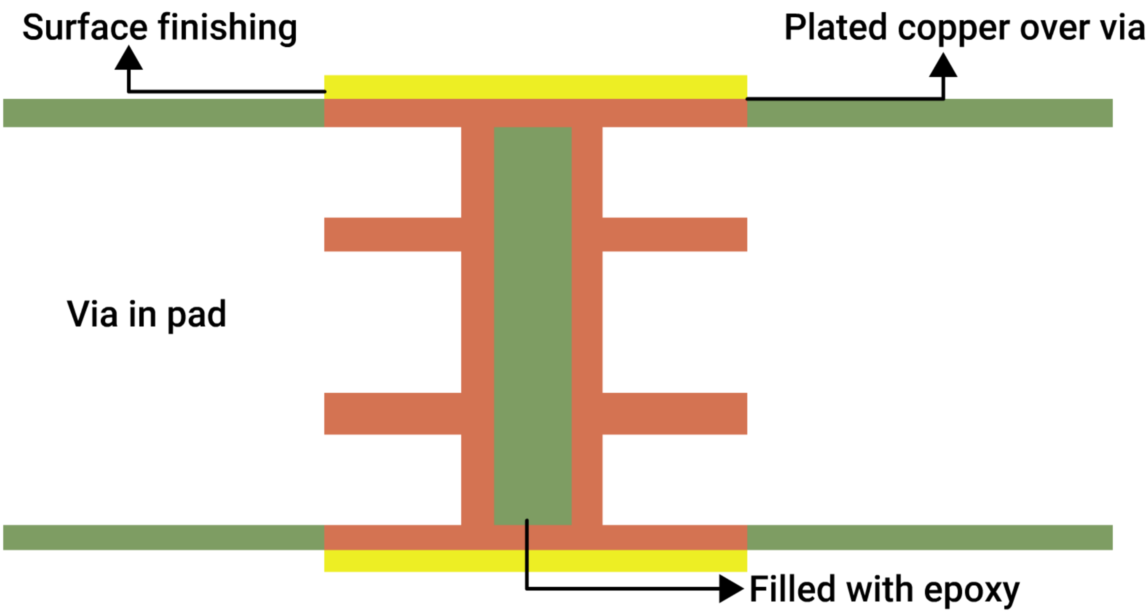
Efficient via placement simplifies manufacturing, lowers fabrication costs, reduces layer count, and minimizes thermal stress risk.

4.1.3 Via fill and capping methods

In HDI and high-speed PCB designs, via filling and capping are essential for creating a flat, solderable surface and improving thermal and electrical reliability.

There are multiple via fill methods available depending on design requirements, cost, and class of reliability.

Types of via fill materials and methods



An illustration of a via filled with non-conductive epoxy

Type	Filling material	Typical use case	IPC class suitability
Conductive fill	Electroplated copper	Via-in-pad, thermal vias, RF/high-speed nets	Class 2, Class 3 (preferred)
Non-conductive epoxy fill	Epoxy	HDI boards where vias are not used for the current path	Class 2, sometimes Class 3
Epoxy resin fill + copper cap	Resin and capped with copper	BGA and fine-pitch QFN pads require flat surfaces	Classes 2 and 3
Copper paste (rare)	Sintered copper paste	Exotic or high-performance RF/ thermal applications	Mostly Class 3

Via cover methods defined by IPC-4761:

1. Open vias: Acceptable in non-critical areas (non-VIP locations).

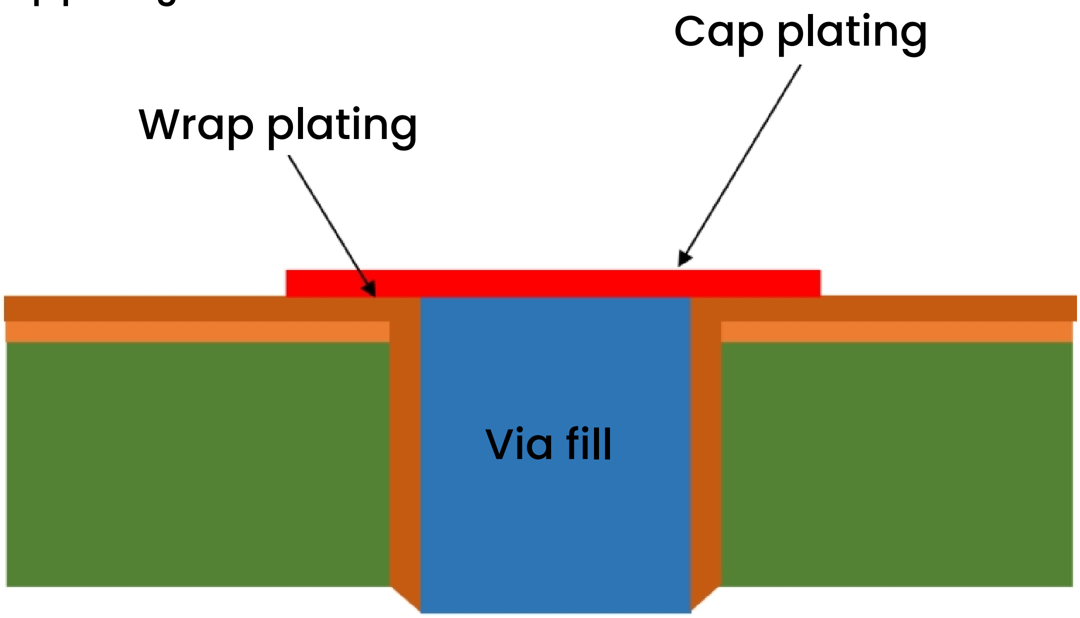
2. Tented vias: Prevent solder wicking; not ideal for thermal/high-current vias.

3. Filled and capped vias: Required for HDI stack-ups and vias needing mechanical/thermal integrity.

For class 3 boards, the cap plating should be a min of 12 µm/0.472 mil, and the protrusion in via can't be more than 50 µm/1.96 mil.

4.1.4 Copper wrap plating, backdrilling, and etchback

Copper wrap plating



An illustration of copper wrap plating

Copper wrap plating involves depositing copper around the via barrel and extending it onto the pad surface. This strengthens:

- Mechanical adhesion
- Current-handling capacity
- Thermal reliability

It's especially important in:

- Aerospace and defense boards (class 3 PCBs)
- HDI boards with sequential lamination

The illustration above shows a cross-section of the via barrel, where electroplated copper extends onto the surface pad, creating a seamless connection between the internal via wall and the pad through wrap plating.

Wrap plating specs	Typical values
Barrel plating thickness	25–40 µm
Surface wrap extension	~50 µm beyond pad edge
Plating uniformity	±5 µm

IPC 6012 defines copper wrap standards for Class 3 boards:

1. Minimum wrap for through holes is 12 µm/0.472 mil.

2. Minimum wrap for microvias (blind and buried) is 12 µm/0.472 mil.

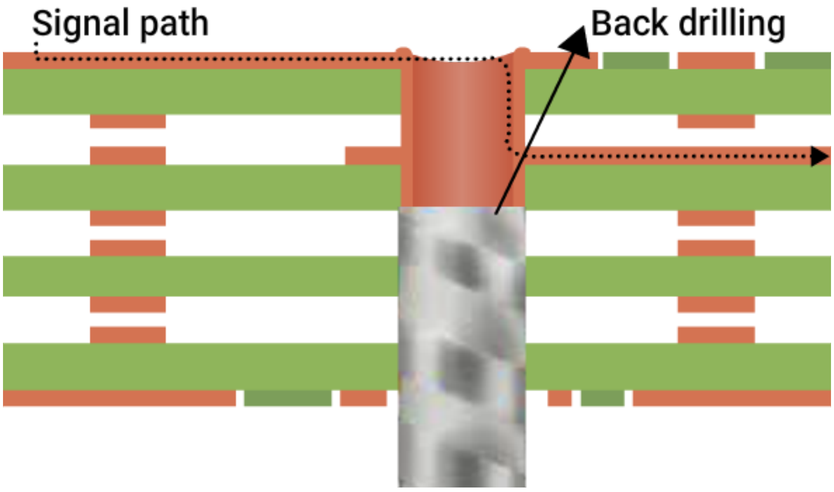
Design tips for efficient copper wrap plating:

1. Ensure proper clearance from other features (min. 10 mil) for reliable wrap plating.

2. Call out "wrap plating required" in the fab drawing.

Backdrilling (Controlled depth drilling)

Backdrilling removes unused via stub portions (the barrel length below the last connected layer), reducing signal reflections and impedance discontinuity.



Backdrilling process to remove the via stub

Consider backdrilling if you're:

- Using interfaces such as SerDes, USB 3.0+, PCIe, SATA, DDR4+.
- Designing boards with data rates >3 Gbps.
- Dealing with via stubs >10 mil.

Important backdrill factors every PCB designer should know:

Parameters	Typical values
Max stub length	<10 mil for 10+ Gbps signals
Backdrill tolerance	±3 mil in depth (varies by shop)
Backdrill annular ring	~12 mil min (design for clearance)

Consider these guidelines if you decide to backdrill vias:

1. Use mechanical layer notes to mark backdrilled vias.

2. Tools like Altium Designer, KiCad, and Allegro support backdrill definitions.

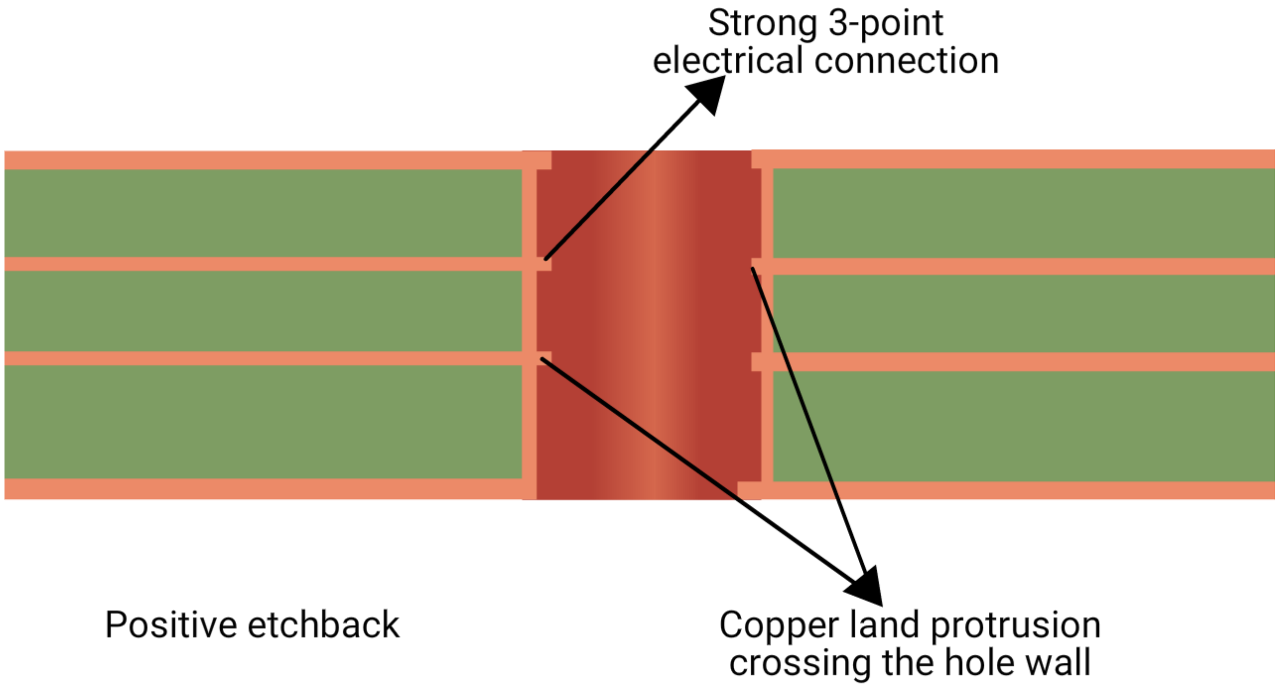
3. Remove non-functional pads to simplify drilling and reduce parasitics.

Etchback

Etchback removes resin smear left after drilling and exposes inner-layer copper before plating. It's critical in multilayer PCBs for strong interconnects.

There are two types:

Negative etchback: Removes more resin than copper.



An illustration of positive etchback

Positive etchback: Removes both resin and copper, creating copper tongues extending into the via barrel.

Benefits of positive etchback:

- Better copper adhesion.
- Improved electrical connectivity.
- Lower risk of delamination.

Positive etchback is commonly used in military, aerospace, medical designs.

Etchback standards:

Typical etchback tolerances	Typical values
Target tongue length for class 3	≥25 μm per layer
Resin removal depth	Controlled based on layer count

Use copper layers with similar thicknesses in your stack-up to ensure uniform etchback. Avoid ultra-thin cores that can over-etch and break.

4.2 Design aspects of via creation

To ensure vias function as intended, you must go beyond placing them for connectivity. It's essential to account for how signals transition through vias and how they interact with nearby copper features and soldermask openings.

This section details critical via design choices and rules to apply in any EDA tool like Altium Designer, KiCad, or Allegro.

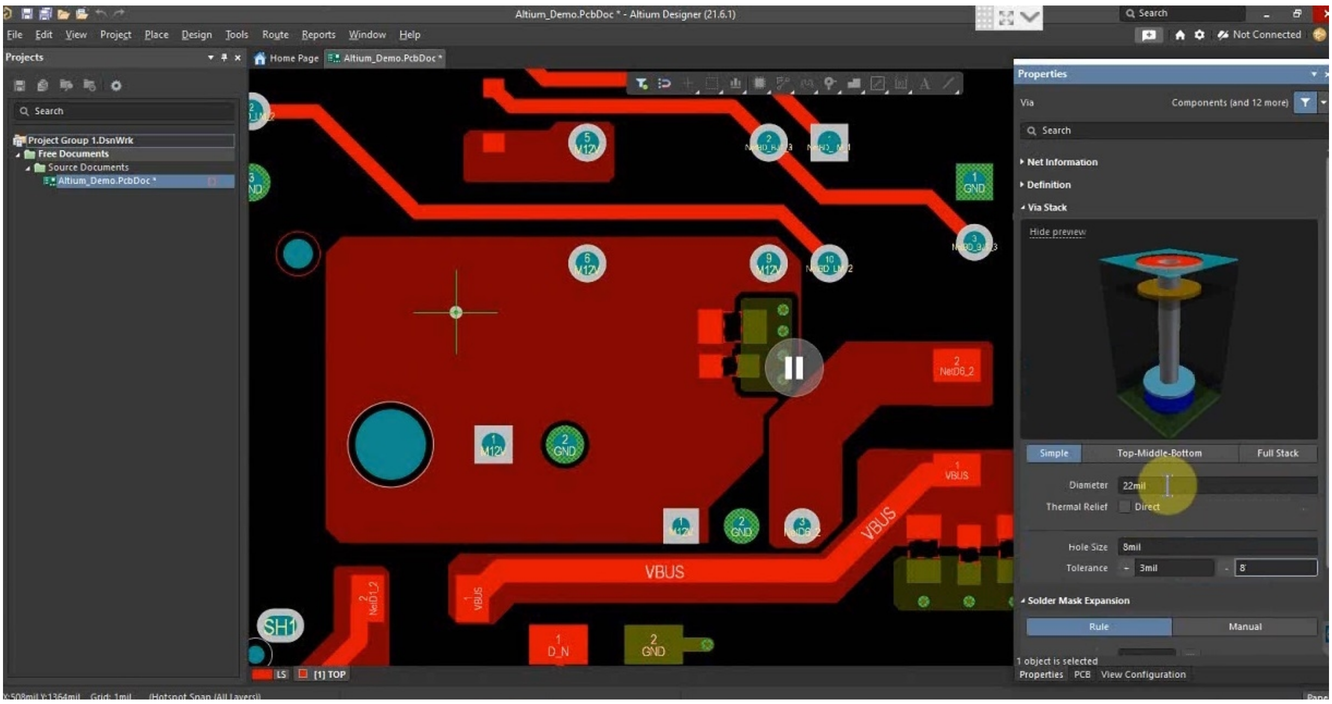
4.2.1 Common via design issues and mitigation techniques

Issue	Cause	Mitigation technique
Wrong via type selection	Using a through-hole via when a blind/buried or microvia is needed	Define multiple via types in your tool's via library, and apply layer-specific via rules
Incorrect via dimensions	Default drill/pad sizes may not align with fab capabilities or impedance needs	Use manufacturer-supported via sizes; validate impedance with field solvers
Unintentional via stubs	Placing through-vias even when the signal transitions between adjacent layers	Define via stacks that span only required layers; use blind/buried via structures where applicable
No return vias for high-speed signals	Not placing ground vias near high-speed signals	Add ground stitching vias close to signal vias (≤0.5 mm), especially for diff pairs and clocks
Vias placed too close	Ignoring spacing rules during auto or manual routing	Define via-to-via spacing rules (≥3× via diameter) in the DRC settings

Issue	Cause	Mitigation technique
Unoptimized pads or antipads	EDA tool defaults may result in high parasitic capacitance	Customize via pad/antipad geometry based on target impedance; validate with impedance tools
Overlapping via-in-pad or trace	Placing VIPs without defining the rules	For via-in-pad, enable VIP rules, use filled and capped via, and specify the requirements in fab notes
Missing documentation for special vias	Designers forget to annotate filled, capped, or backdrilled vias	Add via properties or custom net classes to tag special vias; include exportable notes (e.g., for Gerber/fab) using attributes
Unverified via impedance	No impedance check performed on via + trace combination	Use impedance calculators or simulate with HyperLynx, SIwave, or Altium PDN Analyzer for via transition

4.2.2 Setting via design rules in EDA tools

Modern EDA tools allow you to define via classes, constraints, and layer span rules. These design rules automate DFM compliance and ensure that signal integrity is preserved.



Configuring via properties in an EDA tool

Configuring key via parameters

Parameters	Typical values
Via hole size (finished)	0.2 mm (min), 0.3 mm (standard), 0.4 mm (preferred for through-hole)
Pad size	0.5–0.6 mm for TH vias, 0.35–0.45 mm for microvias
Antipad size	0.8–1.0 mm for 50 Ω, larger for higher-frequency signals
Drill-to-copper clearance	≥8 mil (0.25 mm)
Via-to-via spacing	≥ 6-8mil (standard), ≥10-12 mil for high-speed boards
Aspect ratio (via depth / drill dia)	≤10:1 for PTHs

Always configure separate rules for through, blind, and buried vias based on layer pairs. Define via stacks in EDA tools like Altium's Layer Stack Manager.

Make sure these factors are updated in the design rule settings:

- Minimum via size & pad stack (conforming to IPC-2221B/IPC-2226).
- Layer-specific via spans (e.g., restrict L1–L2 for microvias only).
- Clearance rules between vias and copper features.
- Via-in-pad handling (filled and capped if placed in BGA lands).
- Net class-based via types (e.g., different rules for RF, power, and diff pair nets).

4.2.3 Strategic via placement

Vias introduce parasitic capacitance/inductance and may disrupt return paths, affecting signal integrity.

To avoid these effects, follow below via placement strategies:

Goals	Strategies
Maintain short return paths	Place ground vias near every signal via with a minimum spacing of 8 mil
Minimize stub length	Use backdrilling or blind/buried vias
Avoid dense via clusters	Maintain via-to-via spacing ≥8 mil, especially for high-speed designs
Preserve reference plane continuity	Never place vias over plane splits or cutouts
Avoid skew in diff pairs	Place vias symmetrically and at equal distances

Real-world example

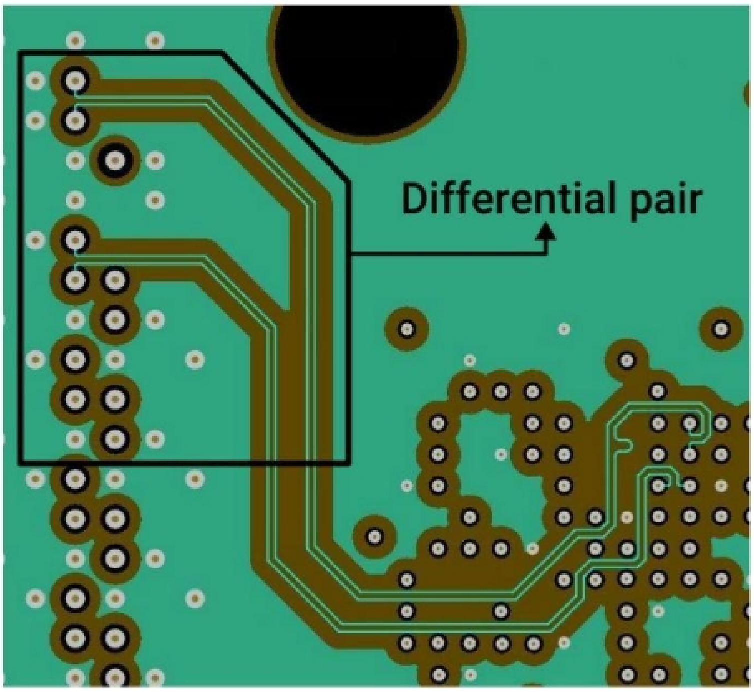
Let's say you have a trace going from layer 1 to layer 6 on a 12-layer board:

If you use a through-hole via, it creates a 6-layer stub, which can resonate. Instead, have a blind via (L1–L6) or a via that terminates at L6 and is backdrilled from the bottom.

Always place ground stitching vias adjacent to your signal vias to contain the return current.

4.2.4 Differential pairs and vias

Differential pair routing is highly sensitive to imbalance, especially at via transitions.



Differential pairs in a PCB layout

Vias can sometimes break the integrity of differential pairs due to:

- Unequal stub lengths leading to timing mismatch.
- Different via environments causing mode conversion.
- Lack of reference vias resulting in field coupling.

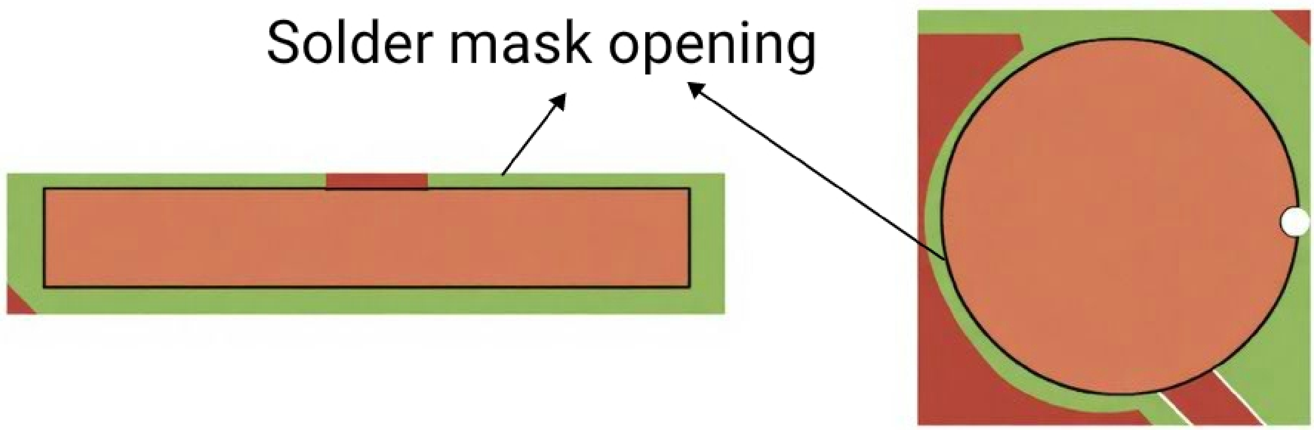
Guidelines for designing vias on differential pairs:

Rules	Significance
Use same drill, pad, antipad sizes	Prevents impedance mismatches
Place vias symmetrically	Keeps length, impedance, and fields balanced
Maintain optimum via spacing (via pair spacing = trace clearance)	Preserves differential impedance
Insert ground via between signal via pair	Reduces crosstalk and improves field containment
Keep stub lengths equal and short	Prevents timing mismatch and reflections

Typical values for a 100Ω-diff pair:

Features	Typical values
Drill diameter	0.3 mm
Pad diameter	0.6 mm
Antipad clearance	1.2 mm
Via-to-via spacing	0.6 mm (center-to-center)
Ground via spacing	≤0.4 mm from each signal via

4.2.5 Soldermask design for vias



Soldermask openings near a via and trace

Soldermask impacts via performance by affecting solder flow, testability, and moisture protection.

Classification of vias with reference to soldermask application:

Type	Description	Use case
Tented	Soldermask covers the via pad entirely	Prevents wicking under BGA/QFN
Untented / exposed	Via remains open for probe or inspection	Test pads, debugging
Tented with a mask plug	Via is filled and covered with soldermask	Additional moisture resistance
Filled and capped (VIPPO)	Via in pad is filled (epoxy or copper) and plated over	Used in high-density BGAs or RF boards

Design rules for soldermask:

Parameters	Typical values
Mask clearance from pad edge	≥1-2 mil
Tenting coverage	Full via + 0.1 mm overlap
Via hole plug depth (mask fill)	≥60% of via height

Via design rule checklist for standard PCBs:

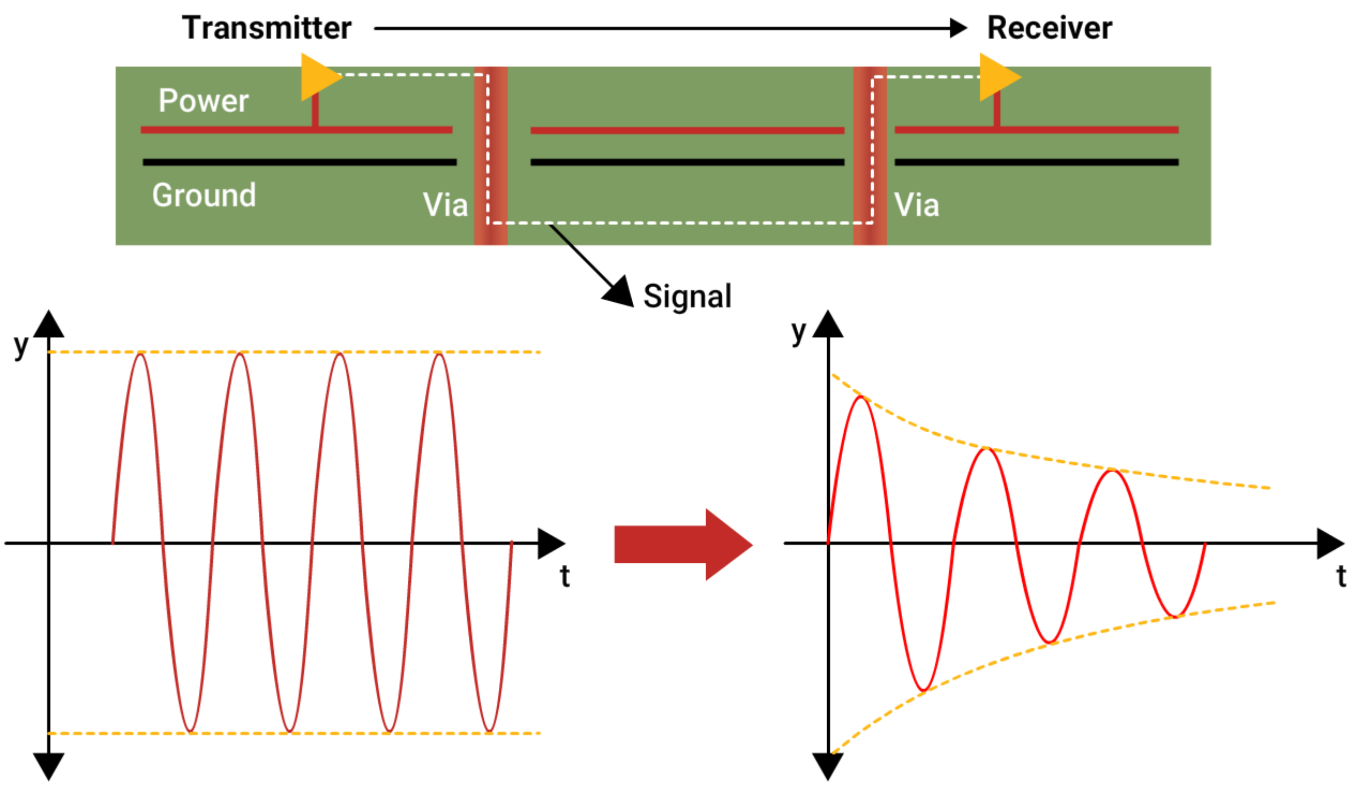
Parameters	Recommended values
Min via hole size	8-12 mil
Max via aspect ratio	≤10:1 (through); ≤0.75:1 (microvia)
Pad size	24 mil for TH; 18 mil for microvia
Antipad clearance	≥1.2× pad diameter
Via-to-via spacing	≥8 mil
Via-to-trace clearance	≥6 mil
Ground stitching via spacing	≤20 mil from signal via
Stub length for high-speed	<12 mil or remove through backdrilling
Differential via skew	<10 ps total
Soldermask clearance	≥2 mil mm beyond the pad
Mask plug depth (tenting)	≥60% via height

5. How vias affect signal integrity

A via impacts signal integrity by introducing impedance discontinuities that can cause reflections, signal loss, and distortion in high-speed circuits. This section explains how via geometry affects transmission characteristics, explores common pitfalls, and provides best practices to mitigate signal degradation.

5.1 Layer transitions and signal degradation

Vias behave like a simple conductor at DC, at high-speed (AC) they introduce parasitics, mainly inductance and capacitance, that can degrade signal quality.



Signal degradation during layer transition

When a signal travels through a via, it experiences a change in geometry, a discontinuity in the reference plane, and often, an abrupt change in trace width or impedance. This results in:

- Reflections due to impedance mismatch.
- Propagation delays due to added inductance.
- Radiation and EMI due to discontinuities.
- Crosstalk if adjacent vias are too close.

A horizontal PCB trace has predictable impedance, typically controlled to 50Ω (single-ended) or 100Ω (differential). But vias introduce uncontrolled vertical impedance.

Often vias span multiple layers, even when the signal only needs to travel between two adjacent ones, leaving via stub that causes signal integrity issues.

Vias can be modelled as an L-C network, it includes:

1. Barrel inductance (L), due to vertical current flow.

2. Pad capacitance (C) , due to pad-to-ground plane overlap.

3. Stub inductance (Ls), due to unused via length (Stub).

Inductance and capacitance of a via is given by:

$$\mathbf{L_{via} \approx 1.27 \times h \times (\ln(4h/d) + 1)\ nH}$$

$$C_{pad} \approx 0.55\ \text{pF (typical for 0.5 mm pad)}$$

Where:

h = via height (mm)

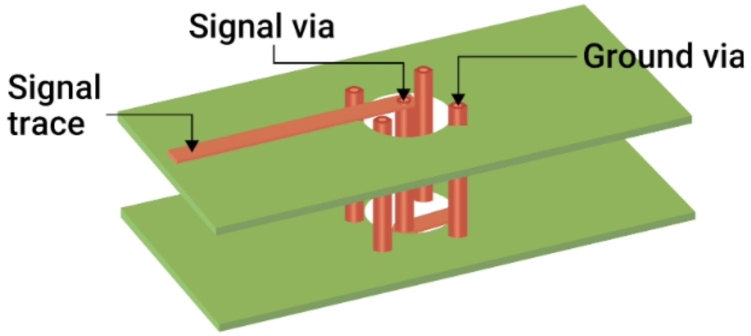
d = drill diameter (mm)

These parasitics become non-negligible above 1 GHz.

For instance, a via of 0.3 mm drill size and 1.6 mm height with L ≈ 1.1 nH, signal degradation begins at f = 1 / (2πL) ≈ 145 MHz.

But reflections and insertion loss become significant above 2–3 GHz, especially if the stub isn't removed.

Via design tips to reduce signal degradation:



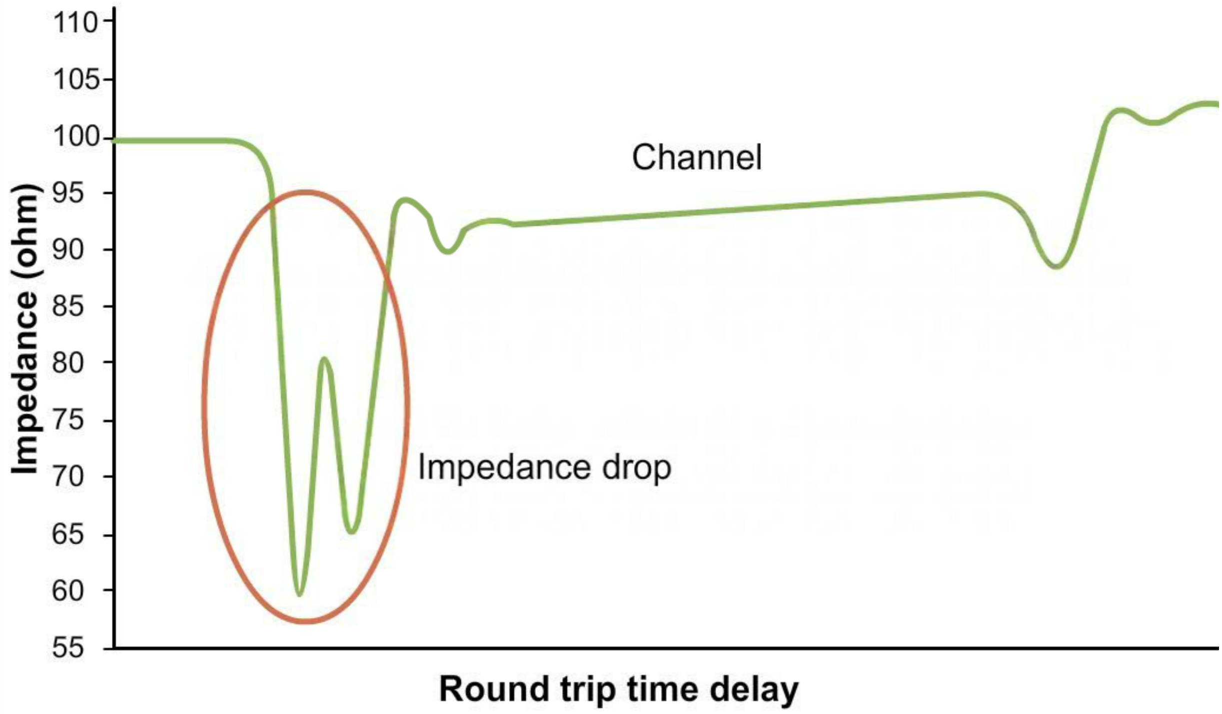
Placing ground vias around a signal via

1. Use vias with diameters between 8–16 mil to limit inductance, and in turn, maintain consistent impedance.

2. Include ground return vias adjacent to signal vias, especially at <3 GHz, to maintain controlled impedance through layer changes.

3. In the 3–5 GHz region and above, denser ground stitching or via fencing reduces impedance spike at the transition.

5.2 Signal reflection and impedance discontinuity



Impedance drop calculated using a TDR

The characteristic impedance (Z_0) of a via is usually lower than that of the trace (e.g., 50 Ω), which causes signal reflection and ringing.

The table below summarizes how vias impact impedance:

Via features	Impact on impedance
Barrel	Conductive plated cylinder, contributes inductance
Pads	Copper rings at each via end, contribute capacitance
Antipads	Clearance holes in reference planes, also affect capacitance
Stub	Unused barrel length reflects signals at certain frequencies

Each of the above via features adds to the overall impedance mismatch between the via and trace.

Consider the following example of calculating the reflection coefficient due to impedance drop.

Trace impedance: 50Ω

Via impedance (with large pad, small antipad): ~30–35Ω

Reflection coefficient $\Gamma = (Z_{via} - Z_{trace}) / (Z_{via} + Z_{trace})$

$\rightarrow \Gamma \approx (35 - 50) / (35 + 50) = -0.176$

This translates to:

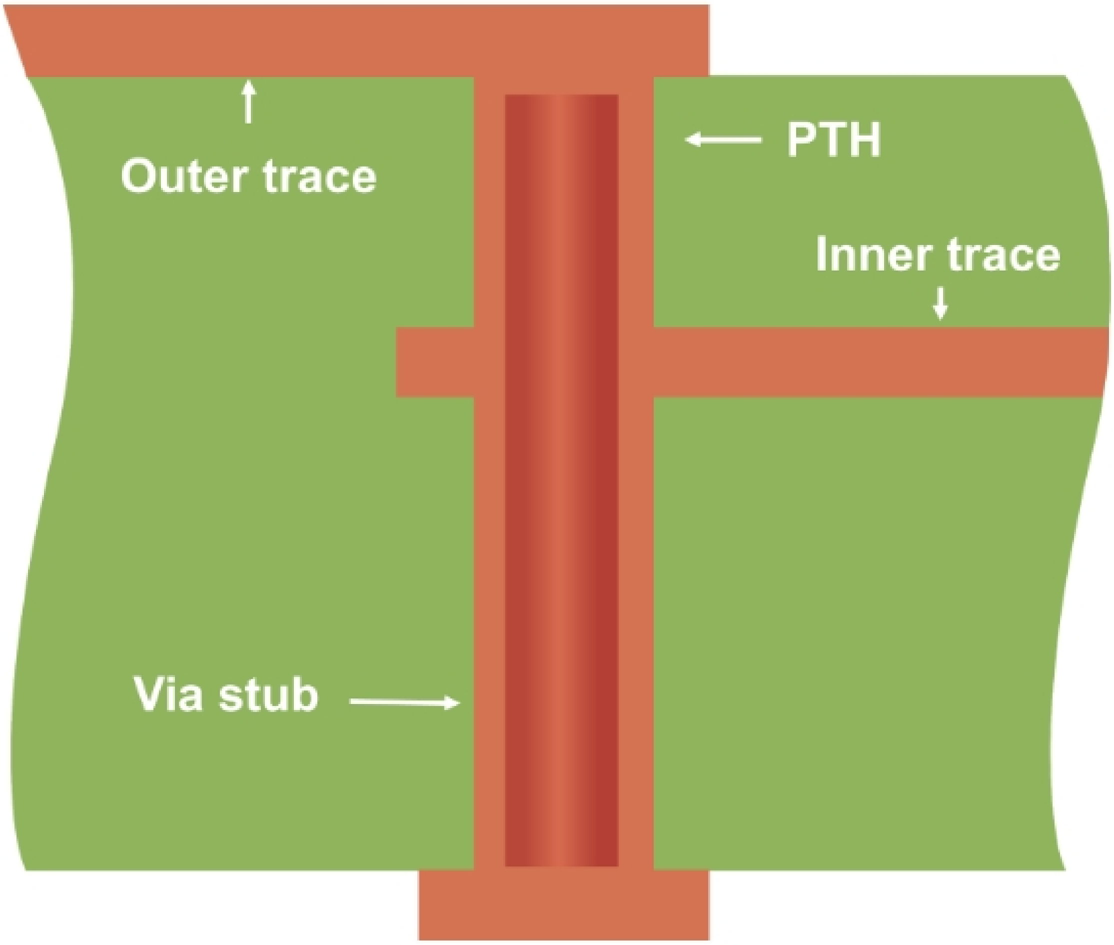
Negative reflection of ~18% of the signal.

Reduced signal swing at the receiver and potential data eye closure and timing violations.

To avoid this, follow the rules given in the table below:

Parameters	Design guidelines
Pad size	Keep as small as possible without violating the design rules
Antipad	Use larger antipads to reduce parasitic capacitance
Barrel height	Minimize via height or stub length
Impedance matching	Match via impedance to trace using 3D EM simulation
Via diameter	Target 0.2–0.4 mm range for lower parasitics
Stitching via spacing	Place stitching vias at least λ/20 apart to minimize inductive variation

5.3 Effects of via stubs



An illustration of a via stub

Via stubs act like open-ended transmission lines, reflecting energy toward the source. The effect is resonant and frequency-dependent, often introducing nulls or notches in the signal path.

Stub resonance frequency is given by the formula below:

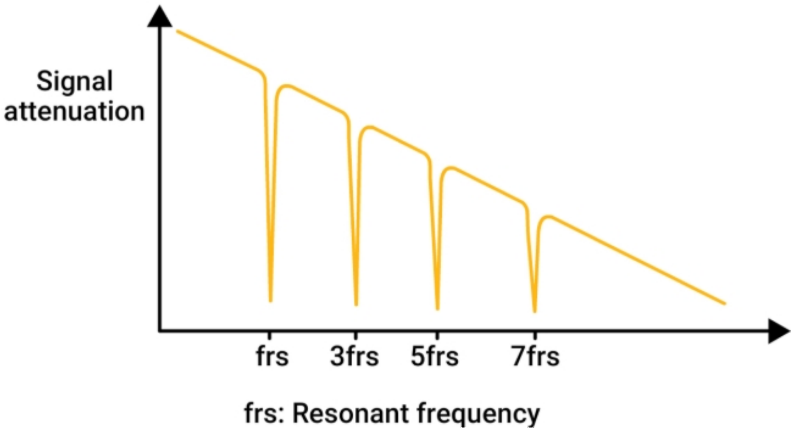
$$f_{res} = \frac{c}{4\sqrt{\varepsilon_r} L_{stub}}$$

Where:

$c = 3\times10^8$ m/s (speed of light)

ε_r = relative permittivity (~4.3 for FR4)

L_{stub} = stub length (in meters)



Signal attenuation at different resonance frequencies

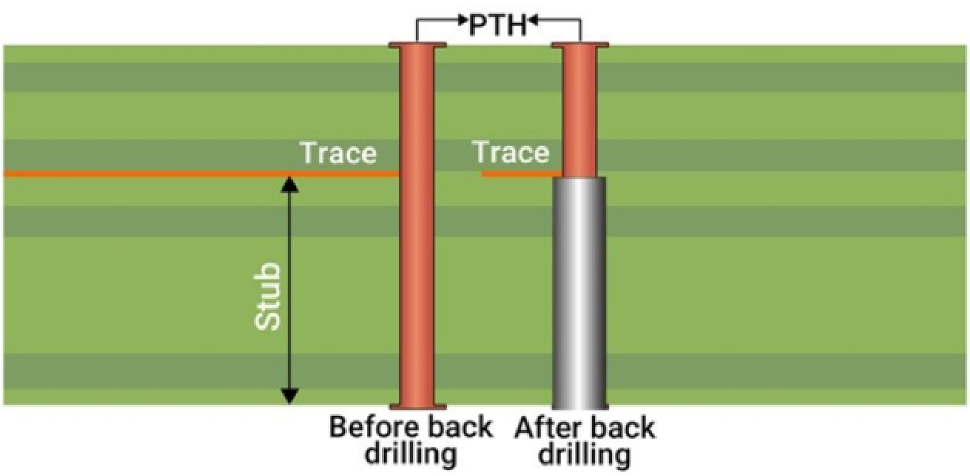
Example:

For FR-4 ($\varepsilon_r \approx 4.3$) and a 0.8 mm stub, $f_{(res)} \approx 4.6$ GHz (right in the operating range of USB 3.0, PCIe Gen3, HDMI 2.1, etc.)

At 4.6 GHz, this stub will resonate and cause signal degradation, ringing, jitter, or eye closure in time-domain signals.

Impact of stubs based on its length:

Stub length	Resonant frequency	Signal impact
0.2 mm	~17 GHz	Minimal for <10 GHz
0.6 mm	~5.8 GHz	Major reflection for RF/DDR
1.0 mm	~3.5 GHz	Severe for USB 3.0 and PCIe

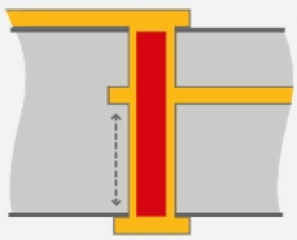


Via stub before and after the backdrilling process

Methods	How it helps	Suitable for
Backdrilling	Removes the unused via section	Designs with through holes
Blind/buried vias	Avoids stubs by connecting only the desired layers	HDI designs
Stub-less vias	Eliminate stubs entirely by controlling layer-to-layer connections	Advanced stack-ups

Maximum via stub lengths for various interfaces:

Applications	Max stub length
USB 3.0 (5 Gbps)	≤0.3 mm
PCIe Gen 3 (8 Gbps)	≤0.25 mm
10G Ethernet	≤0.2 mm
DDR4 / DDR5	≤0.3 mm



PCB DESIGN TOOL

Maximum Via Stub Length Calculator

TRY TOOL

5.4 Via parasitic inductance and capacitance

Vias behave like lumped passive components at high frequencies. Their geometry and proximity to reference planes contribute to parasitic inductance and capacitance, which can disrupt signal transmission and power delivery integrity.

What causes parasitic inductance?

Via barrels form a vertical conductor with a current loop area between the signal and return path. The larger the current loop (i.e., the farther the return path), the higher the inductance.

Via inductance estimation:

$$L_{via} = 1.27 \text{ x } h \text{ x} \left[\ln \frac{4h}{d} + 1 \right] (\text{nH})$$

Where:

h = via height (mm)

d = drill diameter (mm)

Example:

A 0.3 mm via across a 1.6 mm PCB:

$$L_{via} = 1.27 \text{ x } 1.6 \text{ x} \left[\ln \left(\frac{4 \text{ x } 1.6}{0.3} \right) + 1 \right] = 1.15 \text{ nH}$$

At 5 GHz:

$$X_L = 2\pi f L = 2\pi \text{ x } 5 \text{ x } 10^9 \text{ x } 1.15 \text{ x } 10^{-9} = 36\Omega$$

This parasitic impedance of 36Ω causes substantial impedance spike, degrading return loss, and possibly affecting differential balance.

What causes parasitic capacitance?

Parasitic capacitance arises primarily between:

Via pad and ground plane

Via barrel and nearby reference copper

Approximate capacitance:

$$C = 1.41 \text{ x } \varepsilon_r \text{ x } \frac{d^2_{pad}}{d_{anti} \text{ x } h} (pF)$$

Where:

d_{pad} = via pad diameter

d_{anti} = antipad diameter

h = distance to plane

Larger pads and tighter antipads result in higher capacitance and lower via impedance.

The table below summarizes the effects of parasitics on signal integrity:

Parameters	Effect
Inductance	Causes impedance spikes, reflection, EMI
Capacitance	Reduces impedance and causes overshoot/undershoot
Combined L–C	Creates a resonant notch or a frequency dip

Here's how you can reduce parasitics in your via design:

Techniques	What it does
Minimize via length	Reduce inductive path
Increase antipad clearance	Reduce parasitic C
Use ground return vias	Tightens loop, reduces L
Eleminate via stubs	Eliminates resonant behavior
Avoid non-functional pads	Reduces C around signal vias

5.5 Crosstalk between adjacent vias

Crosstalk is unwanted coupling between signals due to electromagnetic field overlap. When signal vias are placed close together, their fields interact, leading to noise injection, timing errors, and signal integrity degradation.

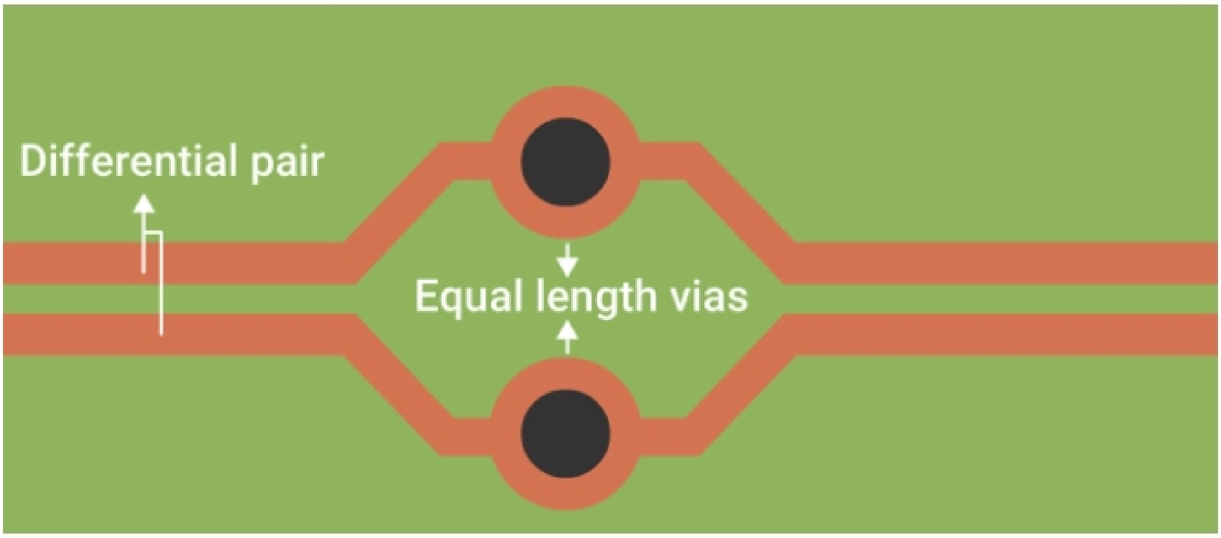
At high frequencies, vias radiate fields that can induce current in nearby vias. Shared return paths or a lack of ground isolation intensifies coupling. Via crosstalk peaks at dense BGA breakout regions or tight diff pair zones.

Two types of via coupling can occur:

Type of coupling	Behavior
Capacitive	High-frequency voltage changes couple between adjacent pads
Inductive	A current loop induces a voltage in nearby loops, especially in diff pairs

The table below shows the factors that impact via coupling:

Factors	Effect
Center-to-center spacing	Primary determinant of capacitive and inductive coupling
Layer depth	Deeper vias affect more internal layers and fields
Return via placement	Reduces coupling by guiding return currents efficiently
Stub presence	Increases coupling through resonances



Designing vias on differential pairs

Differential signals rely on balanced current flow. In a via pair, if one makes more noise than the other:

- Mode conversion occurs (differential ↔ common).
- Skew and jitter increases.
- Eye diagram degrades.

Crosstalk mitigation techniques in vias

Solution	Description
Maintain ≥3× via diameter spacing	Reduces mutual capacitance and inductance
Use ground vias between signal vias	Acts as EM field sink, reduces coupling
Route differential vias symmetrically	Maintains timing and field balance
Avoid shared stubs between nets	Resonant coupling paths form easily

Differential pairs are generally used to route DDR and SerDes signals. Consider these routing tips to avoid coupling:

- Place return vias next to each signal via.
- Avoid fanout patterns that force ground vias away.
- In BGA escape routing, switch to via-in-pad if there is no room for clearances.

Design simulation and verification tools

Tool	Capability
HyperLynx	SI/PI simulation, via models
Ansys SIwave	Full EM simulation of vias
Keysight ADS	S-parameter, eye diagram
Altium Designer/Cadence Allegro	Via modeling with stackup
TDR oscilloscope	Real-time impedance profiling

6. Testing and inspection methods for vias

High-reliability PCB designs rely heavily on robust via structures. Testing methods ensure that vias meet reliability standards throughout fabrication.

As a designer, understanding these testing methodologies and compliance classes is essential for specifying acceptable quality and ensuring your product is manufactured as intended.

6.1 Cross-sectional analysis

Cross-sectioning (or microsectioning) is a destructive testing method used to inspect the internal structure of vias and verify plating, registration, and adhesion quality. It's the most direct and informative way to verify the integrity of multilayer interconnects.

The table below summarizes the features examined during cross-sectional analysis and their significance:

Features evaluated	Why it matters
Plating thickness	Ensures adequate current carrying and fatigue strength
Barrel integrity	Checks for cracks, voids, or delamination
Inner layer connection quality	Confirms the bond between via barrel and the pad
Annular ring concentricity	Reveals registration issues
Resin smear or etch-back quality	Affects long-term conductivity and heat dissipation

If your design has HDI layers, buried/blind vias, or sequential lamination, request for cross-section coupons in the panel design. These coupons help evaluate process consistency without destroying the actual board.

An example of a board cross-section showcasing a via's internal structure

IPC-TM-650 2.1.1 and 2.1.1.2 provide the guidelines for the microsectioning technique and visual acceptance criteria.

An overview of the cross-sectional analysis:

1. An analyst selects a representative PCB from a production run, typically a coupon or a real board selected randomly or based on concern.

2. The technician identifies and isolates the critical test areas, such as via-in-pad structures, blind/buried vias, or fine-pitch interconnects.

3. The board is cut perpendicular to the panel surface (across the via axis) using a precision saw to expose the cross-sectional profile of the vias.

4. The cut section is mounted on a cross-section holder, typically using epoxy or thermosetting resin to keep it stable during grinding.

5. The sample is polished using abrasive silicon carbide or diamond-embedded papers in a progressive grit series (e.g., 320 → 600 → 1200) until a smooth, flat surface is achieved.

6. The polished section is examined under a metallurgical microscope, typically at 100x–200x magnification. The operator checks for:

- Plating thickness uniformity.
- Voids in copper barrels.
- Resin smear or delamination.
- Etchback or over-etching.
- Cracks or interconnect separation.

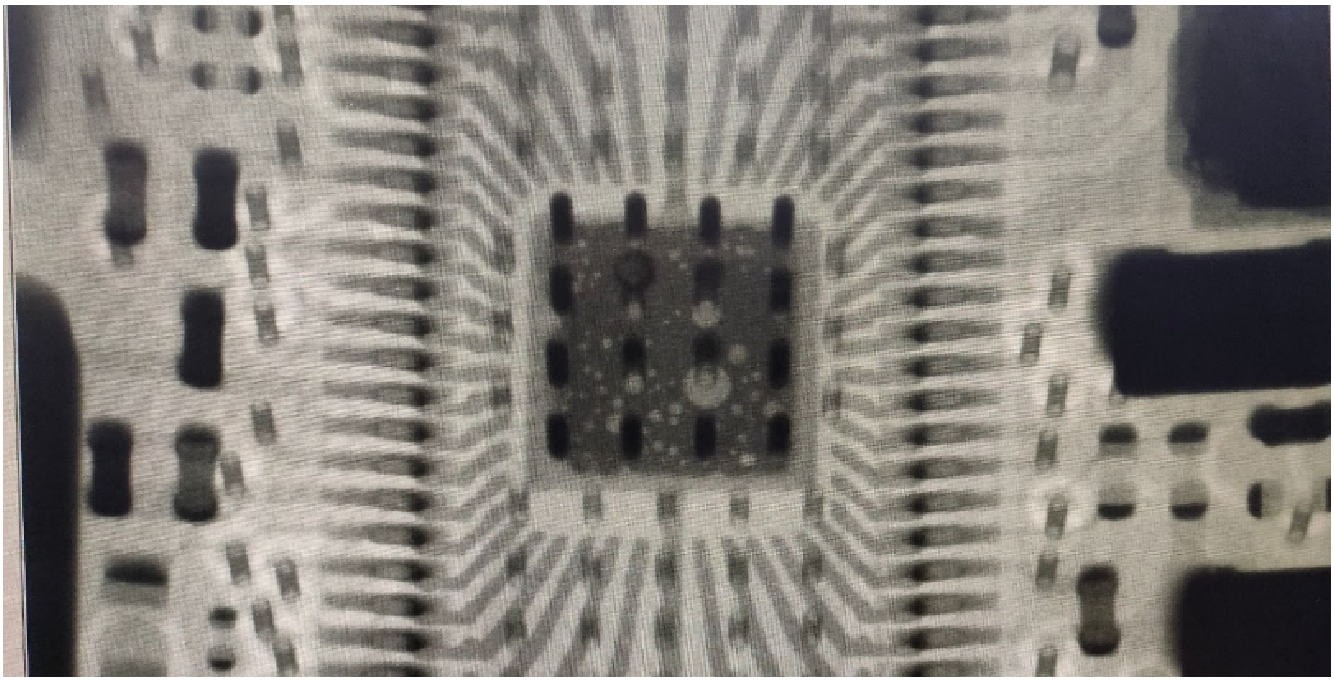
7. High-resolution digital images are captured at appropriate magnification (usually 100x) and archived. These images are used for comparison against IPC acceptance criteria.

This procedure ensures the manufacturer meets structural integrity requirements for vias, especially when the board is intended for IPC class 3 (high-reliability) applications. As a designer, requesting microsection reports during prototyping and lot qualification builds confidence in the long-term reliability of your PCB interconnects.

6.2 X-ray inspection

X-ray inspection is a non-destructive method particularly useful for evaluating via-in-pad structures and BGA solder joints.

Errors detected by X-ray inspection	Why it matters
Misaligned vias	Early detection of registration problems
Barrel voids or incomplete fill	Helps verify via fill quality for VIP structures
Cold solder joints in via-in-pad designs	Prevents reflow or field failure
Internal layer short/open detection	Enables early rejection of faulty boards



A 2D X-ray imaging of a PCB

An overview of X-ray inspection procedure

1. The PCB is placed flat on the X-ray machine platform. No mechanical modification is needed, making this method suitable for finished or in-process boards.

2. The operator calibrates the X-ray source and detector for proper penetration depth and resolution, based on board thickness and density.

3. Specific areas of interest, typically via-in-pad areas, BGA footprints, or high-density interconnect zones, are selected for analysis.

4. For a 2D X-ray, a high-contrast image is taken to show density changes. For 3D (CT or AXI), the system rotates and captures layered images to reconstruct an internal view.

5. The technician or software algorithm checks for:

- Barrel voids in filled vias.
- Misregistration between the drill and the pad.
- Solder voids or bridges under VIP areas.
- Misaligned stacked vias.
- Inadequate via fill or resin gaps.

6. Faulty areas are annotated and exported as part of the inspection report. Resolution typically ranges from 5–25 microns.

When using via-in-pad, especially under BGA components with fine pitch (e.g., 0.4 mm), ask for X-ray verification during first article inspection (FAI) or production start.

In the fab notes, specify AXI (automated X-ray inspection) for boards with BGA with VIP, or when void-free via fill is required in IPC class 3 applications.

6.3 Thermal cycling test

Repeated thermal stress, due to power cycling or environmental exposure, causes copper fatigue in via barrels. This can lead to microcracks, reduced conductivity, and even full via failure.

IPC-9701 provides guidelines on thermal cycling test procedures for assessing interconnect reliability. Typical stress tests include –40°C to +125°C, 1000 cycles.

The following table shows defects observed during thermal cycling tests:

Defects	Causes	Consequences
Barrel cracking	Expansion mismatch between copper and laminate	Open circuits, intermittent failures
Corner cracking	High z-axis expansion stress at pad junctions	Increased impedance or total discontinuity
Delamination	Resin separation at copper-dielectric interface	Reduces mechanical strength, reliability

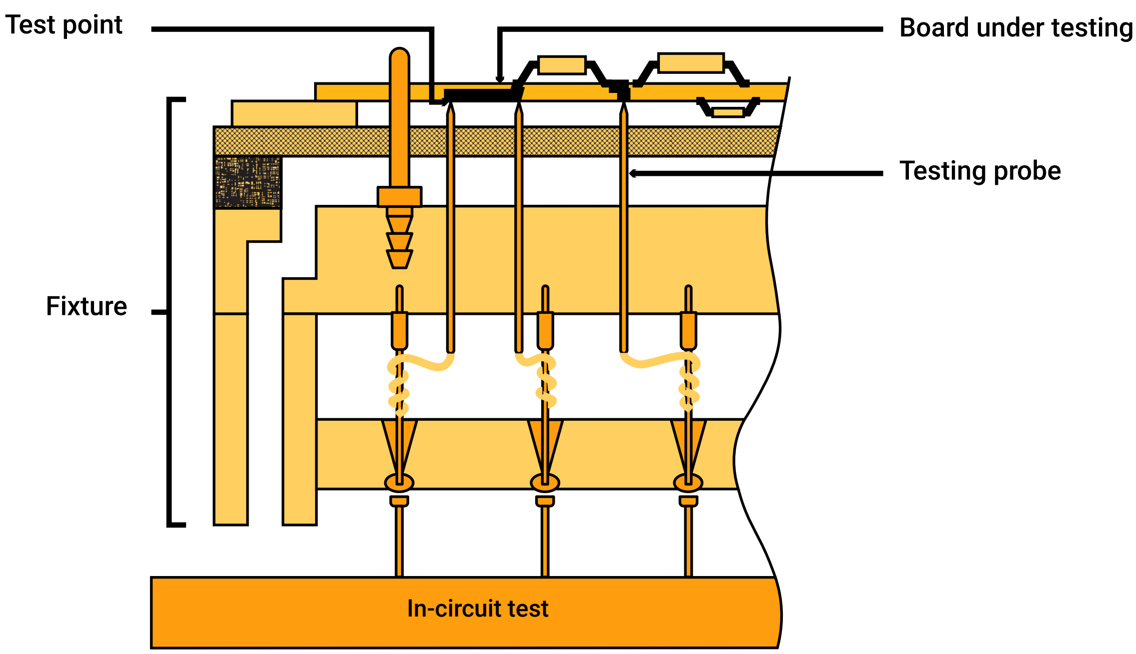
The table below summarizes the causes of thermal fatigue:

Causes	Effects of fatigue
High aspect ratio vias	More susceptible to cracking
Thin plating (<18 μm)	Lower endurance to thermal cycles
Poor resin removal	Increases mechanical stress on the barrel

Specify acceptable thermal reliability and test required in the fab notes, especially for IPC class 3 mission-critical boards. Ensure the via structures don't exceed the recommended aspect ratios.

6.4 IST testing for via connectivity

Interconnect stress testing (IST) is a rapid-cycle test method that uses electrical current to heat vias and cycle them thermally. This induces stress at the most vulnerable junctions, typically at the via-to-inner-layer interfaces.



Setting up the board for IST

Overview of the IST testing procedure

- The designer includes a daisy-chained via coupon in the panel layout. The chain connects vias across various layers (e.g., L2–L10) for stress profiling.
- Each coupon is connected to an IST test unit, which supplies electrical current to generate localized heat and continuously monitors resistance across the chain.
- Resistance heating raises internal temperatures to ~150°C–200°C. This simulates internal via stress under power usage. The system runs for 1000–2000 cycles.
- A sharp resistance increase (>10%) flags a potential failure (e.g., barrel crack, interface delamination).
- Time-to-failure, layer location, and failure mode are recorded. Coupons may be cross-sectioned post-test for visual confirmation.

What designers should consider

- IST is most effective when buried or stacked vias are present.
- Design IST coupons for important layer transitions (e.g., L2-L10 in a 12-layer board).
- Ask for IST performance reports during the qualification phase, especially for class 3 products.

7. Predefined fab notes for vias

Proper documentation of via types, drill charts, and layer connections is a critical step in bridging your PCB layout with the manufacturer's production workflow.

Poor or missing fab notes can result in delays, miscommunication, or worse, fabrication errors that compromise the board's functionality or reliability.

Drill charts and stack-up are essential elements of your fabrication output package. They provide a concise visual and tabular summary of all via-related mechanical operations.

7.1 Drill chart best practices

A drill chart should clearly show:

- Tool number/ drill symbol
- Finished hole size (after plating)
- Drill type (PTH, NPTH, blind, buried, backdrill)
- Start and end layers
- Quantity
- Tolerance
- Plating requirement

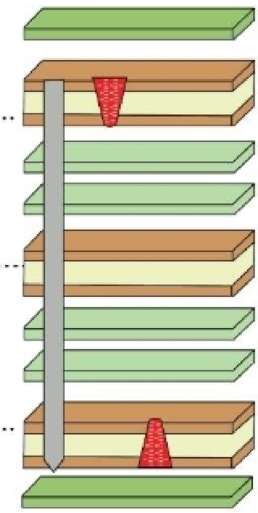
DRILL CHART: TOP to BOTTOM				
ALL UNITS ARE IN MILS				
FIGURE	FINISHED_SIZE	TOLERANCE_DRILL	PLATED	QTY
	10.0	+2.0/-10.0	PLATED	86
	14.0	+2.0/-14.0	PLATED	65
	79.0	+3.0/-3.0	PLATED	2

Example of a drill chart

Drill span and via layer pairs

The stack-up view shows how copper layers, dielectric, and via spans are arranged. Your stack-up should indicate:

- Layer count and names (e.g., L1 to L12)
- Layer usage (signal, plane, mixed)
- Core/prepreg thickness
- Dielectric constants
- Drill spans per via type



Type	Image	Plt (Mil)	Tol (Mil)	Er	Material
Mixed / Powe...		0.7 / 0	0.535	3.48	R04350B-LoPro
			0.36	3.63	FR-370HR
			0.36	3.63	FR-370HR
Mixed / Mixed		0.7 / 0.7	1.05	4.24	FR-370HR
			0.36	3.63	FR-370HR
			0.36	3.63	FR-370HR
Powe... / Mixed		0 / 0.7	0.535	3.48	R04350B-LoPro

An example stack-up with drill representation

Essential parameters to double-check:

- Correct units (mm vs mils).
- Drill sizes after plating vs before plating.
- Whether holes are tented or exposed.
- Via-to-copper clearances (especially for HDI).

Use unique net names for via types in your CAD tool to help generate layer-specific drill files. Keep via span consistent across similar locations to simplify manufacturing.

7.2 Fab notes for via-in-pad, filled vias, and backdrilling

Along with the drill chart and drawing, fab notes should explicitly state which via types are used and their intended layer pairs, especially for HDI or sequentially laminated designs.

Fab note example:

- `All PTH vias span L1-L12: 0.35 mm drill, 0.60 mm pad.`
- `Blind vias L1-L2 and L11-L12: 0.25 mm drill, copper-filled.`
- `Buried vias between L3-L6: 0.20 mm drill.`

Here is how you can indicate other special features of vias in the production documentation:

Via-in-pad (VIP / VIPPO)

Where to specify:

- Mark on silkscreen/mechanical layer.
- Annotate in fabrication notes and assembly drawing.
- Indicate as a VIP or VIPPO in the drill table and include:
    - Drill size
    - Fill type (conductive or non-conductive)
    - Planarization requirement
    - Surface finish

Fab note example:

```
U1, U4, U8 BGA pads have via-in-pad with epoxy fill and planarized
surface. Apply ENIG finish. Do not tent these vias.
```

Via filling:

Where to specify:

- In the drill table, identify which drills require filling.
- In the fab notes, define fill material and type (conductive or non-conductive).
- Use a separate via detail drawing if needed.

Important parameters to declare:

- Fill material (resin/epoxy/copper-filled).
- Planarization required or not.
- Which layers and via types require filling (e.g., all microvias).

Fab note example:

```
All blind vias between L1-L2 and L7-L8 shall be epoxy-filled and
planarized. Use non-conductive fill (e.g., Peters PP2795).
```

Via capping

Where to specify:

- **Fabrication notes:** Identify whether capping is via solder mask or copper plated.
- **Gerber:** The vias which are capped with solder mask should not have a solder mask opening.

Fab note example:

```
VIPs shall be copper capped and planarized for surface mounting. All
other vias shall be mask-capped unless noted.
```

Via tenting

Where to specify:

- **Solder mask layers:** Vias without mask openings are considered as tented vias.
- **Fabrication notes:** Mention the tenting rule based on via size or location.

Fab note example:

```
All vias ≤ 0.3 mm shall be tented with LPI solder mask on both top and
bottom sides.
```

Backdrilling

Where to specify:

- **Drill drawing:** Use unique symbols for backdrills.
- **Fabrication notes:** Define stub removal depth or max stub length.

Fab note example:

```
Backdrill from L1 to L3 for select nets to remove stubs. Target
residual stub length ≤ 10 mils.
```

Etchback

Where to specify:

- Fabrication notes.
- Layer stack-up (optional).

Fab note example:

```
Apply positive etchback per IPC-6012 Class 3. Copper land protrusion
must be visible post-desmear.
```

Wrap plating

Where to specify:

- Fabrication notes.
- May also appear in the stack-up drawing if required.

Fab note example:

```
Wrap plating required on through-hole vias with 360° wrap coverage over
annular ring as per IPC-4761 Type VII.
```

Sample fabrication notes

```
1. Material: FR4, Tg 170°C, UL certified.

2. Board Thickness: 1.6 mm ±10%.

3. Layer Count: 8 Layers (see attached stack-up).

4. Copper Thickness: 1 oz (outer), 0.5 oz (inner).

5. Surface Finish: ENIG (Electroless Nickel Immersion Gold).

6. Solder Mask: LPI Green, both sides.

7. Silkscreen: White, Top and Bottom.
```

VIA DETAILS:

8. Through-hole vias: Plated, 0.30 mm drill size.

9. Blind vias (L1–L2, L7–L8): Laser-drilled, 0.10 mm, epoxy-filled and planarized.

10. Buried vias (L2–L3): Mechanically drilled, no fill.

11. VIPs under U1, U4: Conductive epoxy-filled, copper capped, planarized, ENIG surface.

12. All microvias are staggered, not stacked.

13. Tenting: All vias ≤ 0.3 mm to be tented both sides using LPI solder mask.

14. Backdrill: From L1 to remove stubs beyond L3; max stub length ≤ 10 mils.

15. Fill Material: Taiyo THP-100DX1 or equivalent (non-conductive epoxy).

16. Wrap plating required on through-hole vias with 360° wrap over annular ring.

17. Positive etchback per IPC-6012 class 3.

TESTING & COMPLIANCE:

18. Electrical Testing: 100% netlist verification via IPC-D-356.

19. Impedance Control: ±10% per stack-up table.

20. Include impedance and plating test coupons.

21. Fabrication must meet IPC-6012 Class 3 standards.

Other Resources on PCB Via Design

Articles

- PCB Drilling: The Do's and the Don'ts.
- Annular Ring Explained by a PCB Manufacturer.
- Via-in-Pad in PCB Design Manufacturing.
- How Does Laser Drilling Work in PCBs?.
- Copper Wrap Plating Requirement for PCB Manufacturing.
- Design and Manufacture of Staggered and Stacked Vias.
- Design a Via with Current-Carrying Capacity.
- How PCB Vias Interconnect Circuit Board Layers.
- How Via Stitching Facilitates High-Current PCB Designs.
- Via Filling Techniques Designers Need to Know for PCB Fabrication.
- How to Design Reliable Microvias in Your PCBs.
- 4 Via Design Challenges with Layout Solutions.

Knowledge base

- PCB Via Design.
- Drills and Through-Hole Plating.
- Blind and buried vias.
- Annular Ring Size.

- Plated Half-holes/ Castellated Holes.
- Via-in-Pad.
- Via Covering.
- Annular Ring Manufacturing Issues.
- PCB Backdrilling Process.
- Annular Breakout.
- PCB Via Tenting Design Rules and Fabrication Notes.

Webinars by the industry experts

- Dielectric Anisotropy Implications for Transmission Line Impedance and Via Modelling.
- High-Speed PCB Via Design and Manufacturing.
- Via Design Techniques to Build Reliable PCBs.
- Stacked and Staggered Vias to Optimize PCB Design and Manufacturing.

About Sierra Circuits

Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Talk to our experts

Sierra Circuits helps PCB designers plan it right! Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view. Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.

Our facility

Why Sierra Circuits?

Customer-oriented approach: We collaborate closely with you and tailor solutions to meet your unique design requirements.

Rapid high-tech innovation: We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

Integrated systems: Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages at the beginning of all of our partnerships.

Focus ready: We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits' PCB manufacturing capabilities overview

Rigid PCBs

PCB highlights	Fabrication highlights	Quality management systems
Up to 30 layers	Routed arrays	AS9100D
Board thickness: .005" - .250"	Edge plating	ISO 9001:2015
Max panel size: 21" x 29"	Bevels	ISO 13485:2016
Min trace and space: .002"	Heat sinks	
Solder mask feature tolerance: .001"		
Blind and buried vias		

HDI and microelectronics

High tech in small form	Features	Certified for your industry
Special stack-up help / NPI review	Down to 1.5-mil trace space, 2-mil holes	Mil-spec
Up to 30 layers	Blind vias, buried vias, and other microvia techniques	Automotive
Files evaluated by engineers	Up to 5 sequential lamination cycles	Aerospace and defense
Quote sent by email within a few hours	Laser direct imaging	Medical devices
	via-in-pad technology	

Flex and rigid-flex PCBs

Surface finishes	Standard flex materials	Fabrication highlights	PCB classification
HASL	0.5 mil to 5 mil thick polyimide material	Routed array	IPC 6012, Class 1, 2, and 3
Lead-free HASL	1 mil to 5 mil thick copper clad base material	V score	ISO:9001:2015
OSP (Shikoku F2 and Entek)	Flame retardant laminate	Edge plating	ISO 13485: 2015
ENIG	UL and RoHS compliant	Edge castellation	
Immersion silver		Countersink	
Immersion white tin		Bevel	
Tin nickel		Milling	
Selective gold			

Aerospace and defense PCBs

Capabilities	Documents we provide	Ideal for
Layer count: Up to 30	Certificate of conformance	DOD product designers developing systems for mission critical functions
Minimum board thickness:10 mil	Material specifications	Contract manufacturers who need fast, predictable lead times
Thickness tolerance: 10%	Reflow profile copy (included with first article)	Purchasing teams who require pricing and supply-chin transparency
Minimum trace and space width: 3 mil	Photo requirements	Aerospace engineers looking to shore-up their supply chain troubles
Minimum drill-to-copper clearance: 6 mil	First article inspection report	
Maximum panel size: 21" x 29"	Record of calibrated tools used during manufacturing	
Controlled impedance	FPT, AOI, ionic cleanliness report	
Stacked vias		

RF and microwave PCBs

Capabilities	Certifications
Operating frequency: 10 MHz to 30 GHz	PC Manufacturers Qualification Profile (MQP)
Sodium etch hole treatment process	IPC-A-600
Full-body gold plating up to 120u"	IPC-6012 standards
Soft gold without nickel or ENEPIG surface finishes for enhanced board protection	ISO 9001:2015
Design assistance	NADCAP
	ITAR registered
	RoHS/REACH standards

Sierra Circuits' PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

Full turnkey PCB	Partially consigned assembly	Consigned assembly	Assembly capabilities
Design review	Some parts provided by the designer	Components are provided by the PCB designer	BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components
Bare board fabrication	We will source the rest	Sierra Circuits will fully assemble your PCBs and ship them to you.	DFA
Component procurement	Supply chain resilient	Quick turn times	RoHS, leaded, indium, clean and no clean chemistry
Testing	Quick turnaround time	Fully transparent pricing	Component sizes: 0201, 01005, 08004
Quick turnaround time			Paste in-hole

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

Sierra Circuits' PCB design services

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of ±5%
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

Advanced PCB design tools

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

Sierra Circuits PCB certification and registration

IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunny vale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace, and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D - military and aerospace
- ISO 13485:2016 certificate - medical devices

Mil-Spec certified and DLA approved

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid flex PCBs
- Rigid PCBs

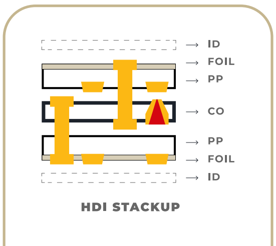
Exceptional material compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

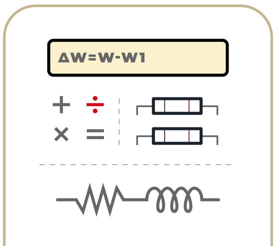
Certified minority supplier by NMSDC

- Minority business certification

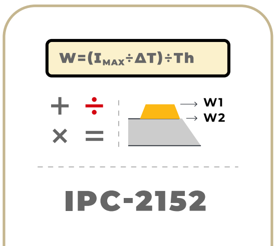
The Designer's Toolkit



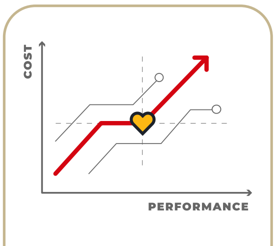
Stackup Designer



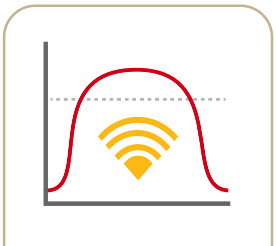
Impedance Calculator



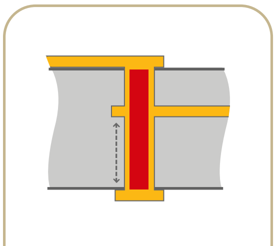
Trace Width, Current Capacity and Temperature Rise Calculator



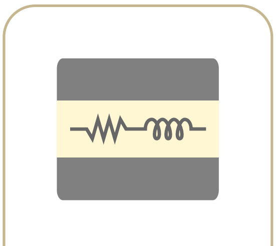
Material Selector



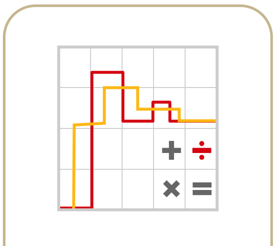
Bandwidth, Rise Time and Critical Length Calculator



Maximum Via Stub Length Calculator

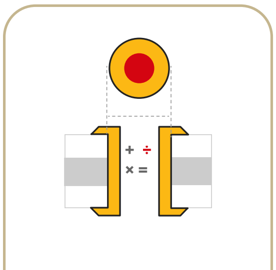


Signal Layer Estimator

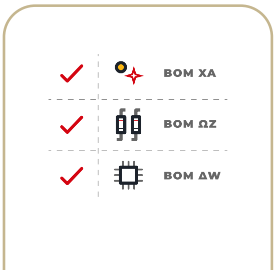


Transmission Line Reflection Calculator

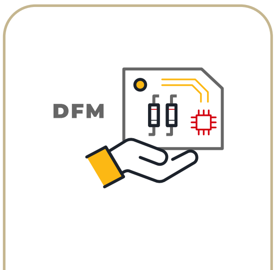
RLC Resonant Frequency & Impedance Calculator



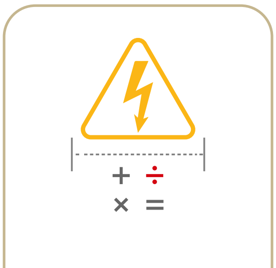
Via Current Capacity and Temperature Rise Calculator



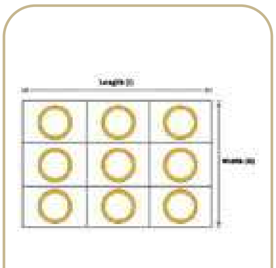
BOM Checker



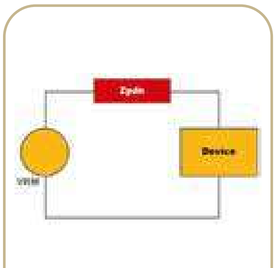
Better DFM



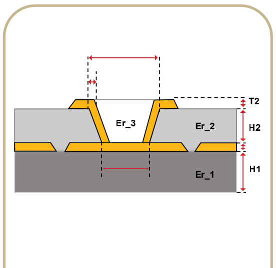
Conductor Spacing and Voltage Calculator



Via Thermal Resisitance Calculator



Power Distribution Network Analyzer



Via Impedance Calculator

Try Tools